

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 2

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1. Linux Server using CLI: Add User and Terminal Switching

a) Download Ubuntu Server 20.04.4 Focal Fossa Virtual Machine image (or any other related version) from Classroom Drive or OSBoxes.com. <https://www.osboxes.org/ubuntu-server/> .

b) Setup the Virtual Machine (VM) in VirtualBox using the downloaded image. Give VM a name as ubuntu-server and use existing virtual hard disk when creating the VM.

c) Login to Virtual Machine (VM) with Username: osboxes and Password: osboxes.org

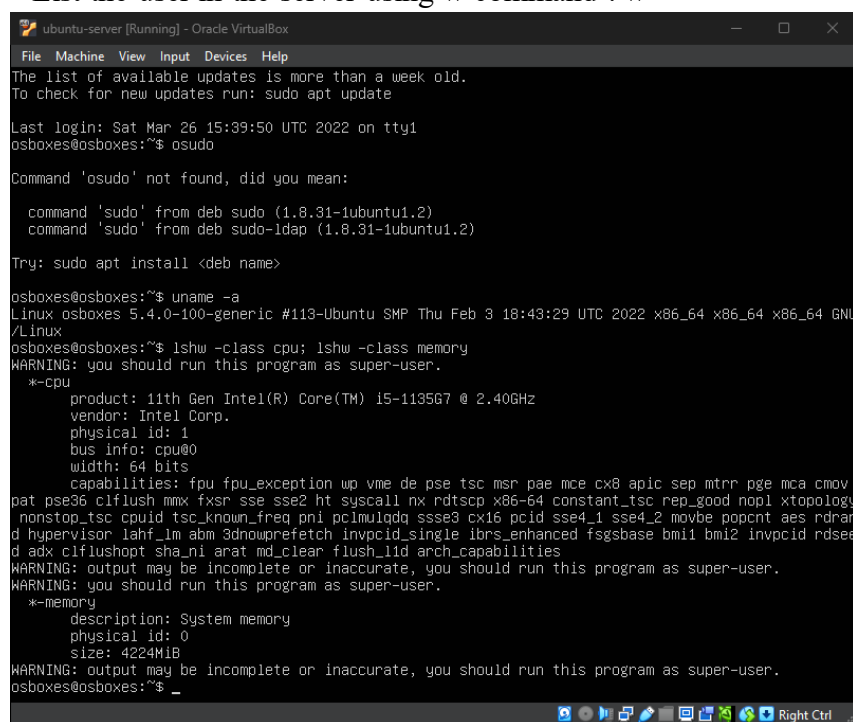
d) Add yourself as a user in the server and give root permission using sudo.

- sudo adduser [username]

e) You can open six terminals with different user by switching to it using CTRL + FN1 ◊ FN6. Switch to terminal 2 by pressing ALT+ F2 and login using your username and password.

f) Use some basic command:

- Print system information using: `uname -a`
- Show CPU and system memory : `lshw -class cpu ; lshw -class memory`
- List the user in the server using `w` command : `w`



```
ubuntu-server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Mar 26 15:39:50 UTC 2022 on tty1
osboxes@osboxes:~$ sudo
Command 'osudo' not found, did you mean:
  command 'sudo' from deb sudo (1.8.31-1ubuntu1.2)
  command 'sudo' from deb sudo-ldap (1.8.31-1ubuntu1.2)

Try: sudo apt install <deb name>

osboxes@osboxes:~$ uname -a
Linux osboxes 5.4.0-100-generic #113-Ubuntu SMP Thu Feb 3 18:43:29 UTC 2022 x86_64 x86_64 x86_64 GNU/Linux
osboxes@osboxes:~$ lshw -class cpu; lshw -class memory
WARNING: you should run this program as super-user.
*-cpu
   product: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz
   vendor: Intel Corp.
   physical id: 1
   bus info: cpu00
   width: 64 bits
   capabilities: fpu fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp x86-64 constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 cx16 pcid sse4_1 sse4_2 movbe popcnt aes rdran
d hypervisorlahf_lm abm 3dnowprefetch invpcid_single ibrs_enhanced fsgsbase bmi1 bmi2 invpcid rdseed
adix cflushopt sha_ni arat md_clear flush_l1d arch_capabilities
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
WARNING: you should run this program as super-user.
*-memory
   description: System memory
   physical id: 0
   size: 4224MiB
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
osboxes@osboxes:~$ _
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
osboxes@osboxes:~$ sudo
Command 'osudo' not found, did you mean:
  command 'sudo' from deb sudo (1.8.31-1ubuntu1.2)
  command 'sudo' from deb sudo-ldap (1.8.31-1ubuntu1.2)
Try: sudo apt install <deb name>

osboxes@osboxes:~$ uname -a
Linux osboxes 5.4.0-100-generic #113-Ubuntu SMP Thu Feb 3 18:43:29 UTC 2022 x86_64 x86_64 x86_64 GNU/Linux
osboxes@osboxes:~$ lshw -class cpu; lshw -class memory
WARNING: you should run this program as super-user.
*-cpu
  product: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz
  vendor: Intel Corp.
  physical id: 1
  bus info: cpu@0
  width: 64 bits
  capabilities: fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp x86-64 constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 cx16 pcid sse4_1 sse4_2 movbe popcnt aes rdprn
d hypervisor lahfm_lbm abm 3dnowprefetch invpcid_single ibrs_enhanced fsgsbase bmi1 bmi2 invpcid rdsee
d adx clflushopt sha_ni arat md_clear flush_l1d arch_capabilities
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
WARNING: you should run this program as super-user.
*-memory
  description: System memory
  physical id: 0
  size: 4224MiB
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
osboxes@osboxes:~$ w
 14:07:18 up 2 min,  1 user,  load average: 0.57, 0.43, 0.18
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
osboxes   tty1    -              14:06    1.00s  0.09s  0.00s  w
osboxes@osboxes:~$ _
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[sudo] password for osboxes:
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,676 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates/main Translation-en [563 kB]
Get:7 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [17.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3,375 kB]
Get:9 http://archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [472 kB]
Get:10 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [548 B]
Get:11 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,238 kB]
Get:12 http://archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [297 kB]
Get:13 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [28.3 kB]
Get:14 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [27.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [7,936 B]
Get:16 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [612 B]
Get:17 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.3 kB]
Get:19 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1,420 B]
Get:20 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Get:23 http://archive.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,302 kB]
Get:24 http://archive.ubuntu.com/ubuntu focal-security/main Translation-en [484 kB]
Get:25 http://archive.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.3 kB]
Get:26 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,247 kB]
Get:27 http://archive.ubuntu.com/ubuntu focal-security/restricted Translation-en [456 kB]
Get:28 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [548 B]
Get:29 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1,016 kB]
Get:30 http://archive.ubuntu.com/ubuntu focal-security/universe Translation-en [214 kB]
Get:31 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [21.4 kB]
Get:32 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [24.8 kB]
Get:33 http://archive.ubuntu.com/ubuntu focal-security/multiverse Translation-en [5,968 B]
Get:34 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [540 B]
Fetched 19.0 MB in 8s (2,488 kB/s)
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Get:4 http://archive.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,676 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates/main Translation-en [563 kB]
Get:7 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [17.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3,375 kB]
Get:9 http://archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [472 kB]
Get:10 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [548 B]
Get:11 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,238 kB]
Get:12 http://archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [297 kB]
Get:13 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [28.3 kB]
Get:14 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [27.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [7,936 B]
Get:16 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [612 B]
Get:17 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.3 kB]
Get:19 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1,420 B]
Get:20 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Get:23 http://archive.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,302 kB]
Get:24 http://archive.ubuntu.com/ubuntu focal-security/main Translation-en [484 kB]
Get:25 http://archive.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.3 kB]
Get:26 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,247 kB]
Get:27 http://archive.ubuntu.com/ubuntu focal-security/restricted Translation-en [456 kB]
Get:28 http://archive.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [548 B]
Get:29 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1,016 kB]
Get:30 http://archive.ubuntu.com/ubuntu focal-security/universe Translation-en [214 kB]
Get:31 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [21.4 kB]
Get:32 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [24.8 kB]
Get:33 http://archive.ubuntu.com/ubuntu focal-security/multiverse Translation-en [5,968 B]
Get:34 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [540 B]
Fetched 19.0 MB in 8s (2,488 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
247 packages can be upgraded. Run 'apt list --upgradable' to see them.
osboxes@osboxes:~$
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
liblvm2 libmtdev libpciaccess0 libpixman-1-0 libsensors-config libsensors5 libsm6 libvulkan1
libwacom-bin libwacom-common libwacom2 libwayland-client0 libwayland-server0 libx11-xcb1
libxatracker2 libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0
libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-util1 libxcb-xf86vm0 libxcomposite1 libxcursor1
libxdamage1 libxfixes3 libxfont2 libfft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4
libxrandr2 libxrender1 libxshmfence1 libxss1 libxt6 libxtst6 libxv1 libxvmc1 libxxf86dga1
libxxf86vm1 mesa-vulkan-drivers x11-common x11-utils x11-xkb-utils xbitmaps xfonts-base
xfonts-encodings xfonts-utils xserver-common xserver-xorg xserver-xorg-input-all
xserver-xorg-input-libinput xserver-xorg-input-wacom xserver-xorg-legacy xserver-xorg-video-all
xserver-xorg-video-amdgpu xserver-xorg-video-ati xserver-xorg-video-fbdev
xserver-xorg-video-intel xserver-xorg-video-nouveau xserver-xorg-video-qxl
xserver-xorg-video-radeon xserver-xorg-video-vesa xserver-xorg-video-vmware
Suggested packages:
lm-sensors mesa-utils xfonts-100dpi | xfonts-75dpi xfonts-scalable xinput firmware-amd-graphics
xserver-xorg-video-r128 xserver-xorg-video-mach64 firmware-misc-nonfree xfonts-cyrillic
The following NEW packages will be installed:
fontconfig-config fonts-dejavu-core libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1
libegl-mesa0 libegl1 libepoxy0 libfontconfig1 libfontenc1 libgbm1 libgl1 libgl1-mesa-dri
libglapi-mesa libglvnd0 libglx-mesa0 libglx0 libice6 libinput-bin libinput10 liblvm2 libmtdev1
libpciaccess0 libpixman-1-0 libsensors-config libsensors5 libsm6 libvulkan1 libwacom-bin
libwacom-common libwacom2 libwayland-client0 libwayland-server0 libx11-xcb1 libxatracker2
libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-shape0
libxcb-shm0 libxcb-sync1 libxcb-util1 libxcb-xf86vm0 libxcomposite1 libxcursor1 libxdamage1
libxfixes3 libxfont2 libfft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4 libxrandr2
libxrender1 libxshmfence1 libxss1 libxt6 libxtst6 libxv1 libxvmc1 libxxf86dga1 libxxf86vm1
mesa-vulkan-drivers x11-common x11-utils x11-xkb-utils xbitmaps xfonts-base xfonts-encodings
xfonts-utils xinit xserver-common xserver-xorg xserver-xorg-core xserver-xorg-input-all
xserver-xorg-input-libinput xserver-xorg-input-wacom xserver-xorg-legacy xserver-xorg-video-all
xserver-xorg-video-amdgpu xserver-xorg-video-ati xserver-xorg-video-fbdev
xserver-xorg-video-intel xserver-xorg-video-nouveau xserver-xorg-video-qxl
xserver-xorg-video-radeon xserver-xorg-video-vesa xserver-xorg-video-vmware xterm
The following packages will be upgraded:
libevdev2
1 upgraded, 95 newly installed, 0 to remove and 246 not upgraded.
Need to get 51.6 MB of archives.
After this operation, 566 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Get:47 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libxpm4 amd64 1:3.5.12-1ubuntu0.20.04.2 [34.9 kB]
Get:48 http://archive.ubuntu.com/ubuntu focal/main amd64 libxaw7 amd64 2:1.0.13-1 [173 kB]
Get:49 http://archive.ubuntu.com/ubuntu focal/main amd64 libxcb-randr0 amd64 1.14-2 [16.3 kB]
Get:50 http://archive.ubuntu.com/ubuntu focal/main amd64 libxcb-shape0 amd64 1.14-2 [5,928 B]
Get:51 http://archive.ubuntu.com/ubuntu focal/main amd64 libxcomposite1 amd64 1:0.4.5-1 [6,976 B]
Get:52 http://archive.ubuntu.com/ubuntu focal/main amd64 libxrender1 amd64 1:0.9.10-1 [18.7 kB]
Get:53 http://archive.ubuntu.com/ubuntu focal/main amd64 libxcursor1 amd64 1:1.2.0-2 [20.1 kB]
Get:54 http://archive.ubuntu.com/ubuntu focal/main amd64 libxdamage1 amd64 1:1.1.5-2 [6,996 B]
Get:55 http://archive.ubuntu.com/ubuntu focal/main amd64 libxfont2 amd64 1:2.0.3-1 [91.7 kB]
Get:56 http://archive.ubuntu.com/ubuntu focal/main amd64 libxft2 amd64 2.3.3-0ubuntu1 [39.2 kB]
Get:57 http://archive.ubuntu.com/ubuntu focal/main amd64 libxi6 amd64 2:1.7.10-0ubuntu1 [29.9 kB]
Get:58 http://archive.ubuntu.com/ubuntu focal/main amd64 libxinerama1 amd64 2:1.1.4-2 [6,904 B]
Get:59 http://archive.ubuntu.com/ubuntu focal/main amd64 libxkbfile1 amd64 1:1.1.0-1 [65.3 kB]
Get:60 http://archive.ubuntu.com/ubuntu focal/main amd64 libxrandr2 amd64 2:1.5.2-0ubuntu1 [18.5 kB]
Get:61 http://archive.ubuntu.com/ubuntu focal/main amd64 libxss1 amd64 1:1.2.3-1 [8,140 B]
Get:62 http://archive.ubuntu.com/ubuntu focal/main amd64 libxtst6 amd64 2:1.2.3-1 [12.8 kB]
Get:63 http://archive.ubuntu.com/ubuntu focal/main amd64 libxv1 amd64 2:1.0.11-1 [10.7 kB]
Get:64 http://archive.ubuntu.com/ubuntu focal/main amd64 libxvmc1 amd64 2:1.0.12-2 [14.2 kB]
Get:65 http://archive.ubuntu.com/ubuntu focal/main amd64 libxxf86dga1 amd64 2:1.1.5-0ubuntu1 [12.0 kB]
Get:66 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 mesa-vulkan-drivers amd64 21.2.6-0ubuntu0.1~20.04.2 [5,788 kB]
Get:67 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libglvnd0 amd64 1.3.2-1~ubuntu0.20.04.2 [48.1 kB]
Get:68 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libglx0 amd64 1.3.2-1~ubuntu0.20.04.2 [32.5 kB]
Get:69 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libgl1 amd64 1.3.2-1~ubuntu0.20.04.2 [85.8 kB]
Get:70 http://archive.ubuntu.com/ubuntu focal/main amd64 x11-utils amd64 7.7+5 [199 kB]
Get:71 http://archive.ubuntu.com/ubuntu focal/main amd64 x11-xkb-utils amd64 7.7+5 [158 kB]
Get:72 http://archive.ubuntu.com/ubuntu focal/main amd64 xbitmaps all 1.1.1-2 [28.1 kB]
Get:73 http://archive.ubuntu.com/ubuntu focal/main amd64 xfonts-encodings all 1:1.0.5-0ubuntu1 [573 kB]
Get:74 http://archive.ubuntu.com/ubuntu focal/main amd64 xfonts-utils amd64 1:7.7+6 [91.5 kB]
Get:75 http://archive.ubuntu.com/ubuntu focal/main amd64 xfonts-base all 1:1.0.5 [5,896 kB]
81% [75 xfonts-base 3,230 B/5,896 kB 0%] 4,786 kB/s 1s
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Preparing to unpack .../63-libxvmc1_2%3a1.0.12-2_amd64.deb ...
Unpacking libxvmc1:amd64 (2:1.0.12-2) ...
Selecting previously unselected package libxxf86dga1:amd64.
Preparing to unpack .../64-libxxf86dga1_2%3a1.1.5-0ubuntu1_amd64.deb ...
Unpacking libxxf86dga1:amd64 (2:1.1.5-0ubuntu1) ...
Selecting previously unselected package mesa-vulkan-drivers:amd64.
Preparing to unpack .../65-mesa-vulkan-drivers_21.2.6-0ubuntu0.1~20.04.2_amd64.deb ...
Unpacking mesa-vulkan-drivers:amd64 (21.2.6-0ubuntu0.1~20.04.2) ...
Selecting previously unselected package libglvnd0:amd64.
Preparing to unpack .../66-libglvnd0_1.3.2-1~ubuntu0.20.04.2_amd64.deb ...
Unpacking libglvnd0:amd64 (1.3.2-1~ubuntu0.20.04.2) ...
Selecting previously unselected package libglx0:amd64.
Preparing to unpack .../67-libglx0_1.3.2-1~ubuntu0.20.04.2_amd64.deb ...
Unpacking libglx0:amd64 (1.3.2-1~ubuntu0.20.04.2) ...
Selecting previously unselected package libgl1:amd64.
Preparing to unpack .../68-libgl1_1.3.2-1~ubuntu0.20.04.2_amd64.deb ...
Unpacking libgl1:amd64 (1.3.2-1~ubuntu0.20.04.2) ...
Selecting previously unselected package x11-utils.
Preparing to unpack .../69-x11-utils_7.7+5_amd64.deb ...
Unpacking x11-utils (7.7+5) ...
Selecting previously unselected package x11-xkb-utils.
Preparing to unpack .../70-x11-xkb-utils_7.7+5_amd64.deb ...
Unpacking x11-xkb-utils (7.7+5) ...
Selecting previously unselected package xbitmaps.
Preparing to unpack .../71-xbitmaps_1.1.1-2_all.deb ...
Unpacking xbitmaps (1.1.1-2) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../72-xfonts-encodings_1%3a1.0.5-0ubuntu1_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu1) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../73-xfonts-utils_1%3a7.7+6_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6) ...
Selecting previously unselected package xfonts-base.
Preparing to unpack .../74-xfonts-base_1%3a1.0.5_all.deb ...
Unpacking xfonts-base (1:1.0.5) ...
Progress: [ 38%] [#####.....]
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Setting up libgl1:amd64 (1.3.2-1ubuntu0.20.04.2) ...
Setting up libxt6:amd64 (1:1.1.5-1) ...
Setting up libwacom-bin (1.3-2ubuntu3) ...
Setting up libinput10:amd64 (1.15.5-1ubuntu0.3) ...
Setting up libxmu6:amd64 (2:1.1.3-0ubuntu1) ...
Setting up libxaw7:amd64 (2:1.0.13-1) ...
Setting up x11-utils (7.7+5) ...
Setting up xterm (353-1ubuntu1.20.04.2) ...
update-alternatives: using /usr/bin/xterm to provide /usr/bin/x-terminal-emulator (x-terminal-emulator) in auto mode
update-alternatives: using /usr/bin/lxterm to provide /usr/bin/x-terminal-emulator (x-terminal-emulator) in auto mode
Setting up x11-xkb-utils (7.7+5) ...
Setting up xserver-common (2:1.20.13-1ubuntu1~20.04.18) ...
Setting up xserver-xorg-legacy (2:1.20.13-1ubuntu1~20.04.18) ...
Setting up xserver-xorg-core (2:1.20.13-1ubuntu1~20.04.18) ...
Setting up xserver-xorg-video-radeon (1:19.1.0-1) ...
Setting up xserver-xorg-input-wacom (1:0.39.0-0ubuntu1) ...
Setting up xserver-xorg-video-fbdev (1:0.5.0-1ubuntu1) ...
Setting up xserver-xorg-video-vmware (1:13.3.0-3) ...
Setting up xserver-xorg-video-intel (2:2.99.917+git20200226-1) ...
Setting up xserver-xorg-video-vesa (1:2.4.0-2) ...
Setting up xserver-xorg-video-qxl (0.1.5+git20200331-1) ...
Setting up xserver-xorg-video-amdgpu (19.1.0-1ubuntu0.1) ...
Setting up xserver-xorg (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-video-nouveau (1:1.0.16-1) ...
Setting up xserver-xorg-input-libinput (0.29.0-1) ...
Setting up xserver-xorg-video-ati (1:19.1.0-1) ...
Setting up xserver-xorg-video-all (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-input-all (1:7.7+19ubuntu14) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
osboxes@osboxes:~$
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
osboxes@osboxes:~$
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Setting up xserver-xorg-video-qxl (0.1.5+git20200331-1) ...
Setting up xserver-xorg-video-amdgpu (19.1.0-1ubuntu0.1) ...
Setting up xserver-xorg (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-video-nouveau (1:1.0.16-1) ...
Setting up xserver-xorg-input-libinput (0.29.0-1) ...
Setting up xserver-xorg-video-ati (1:19.1.0-1) ...
Setting up xserver-xorg-video-all (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-input-all (1:7.7+19ubuntu14) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
osboxes@osboxes:~$ startx
xauth: file /home/osboxes/.Xauthority does not exist

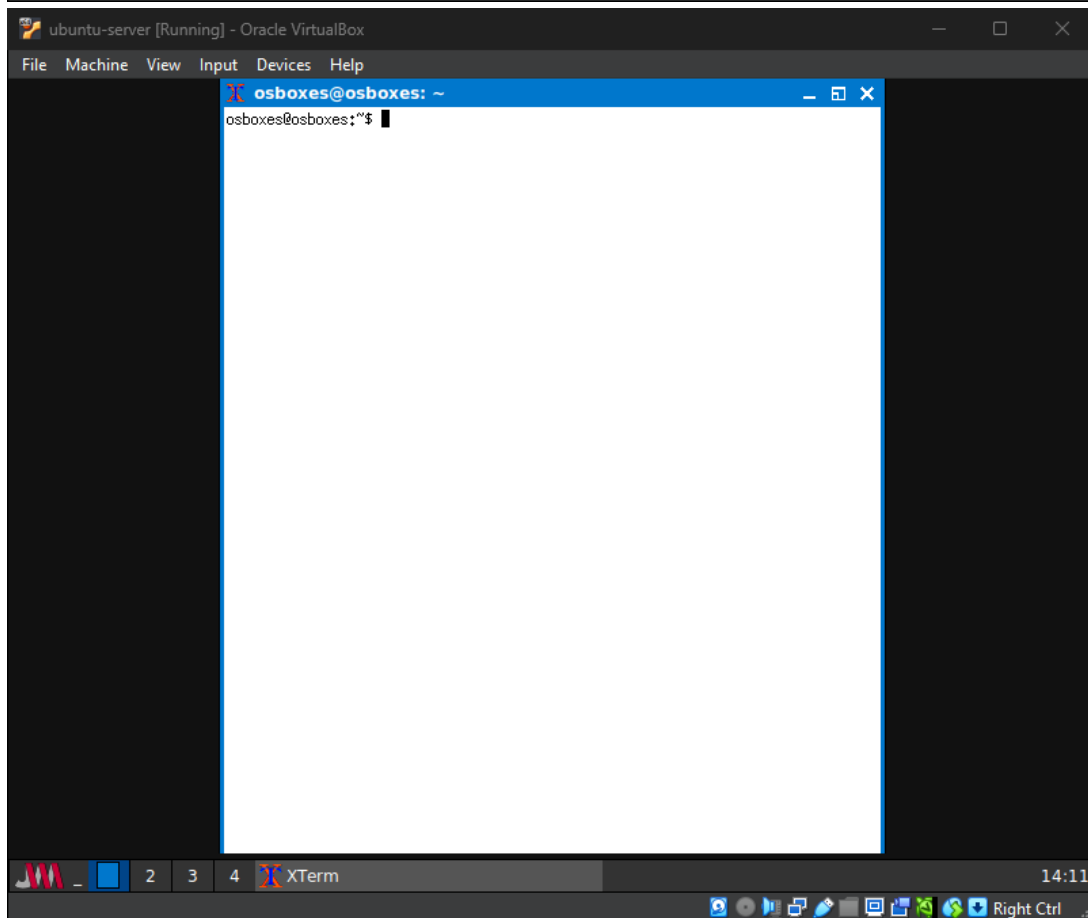
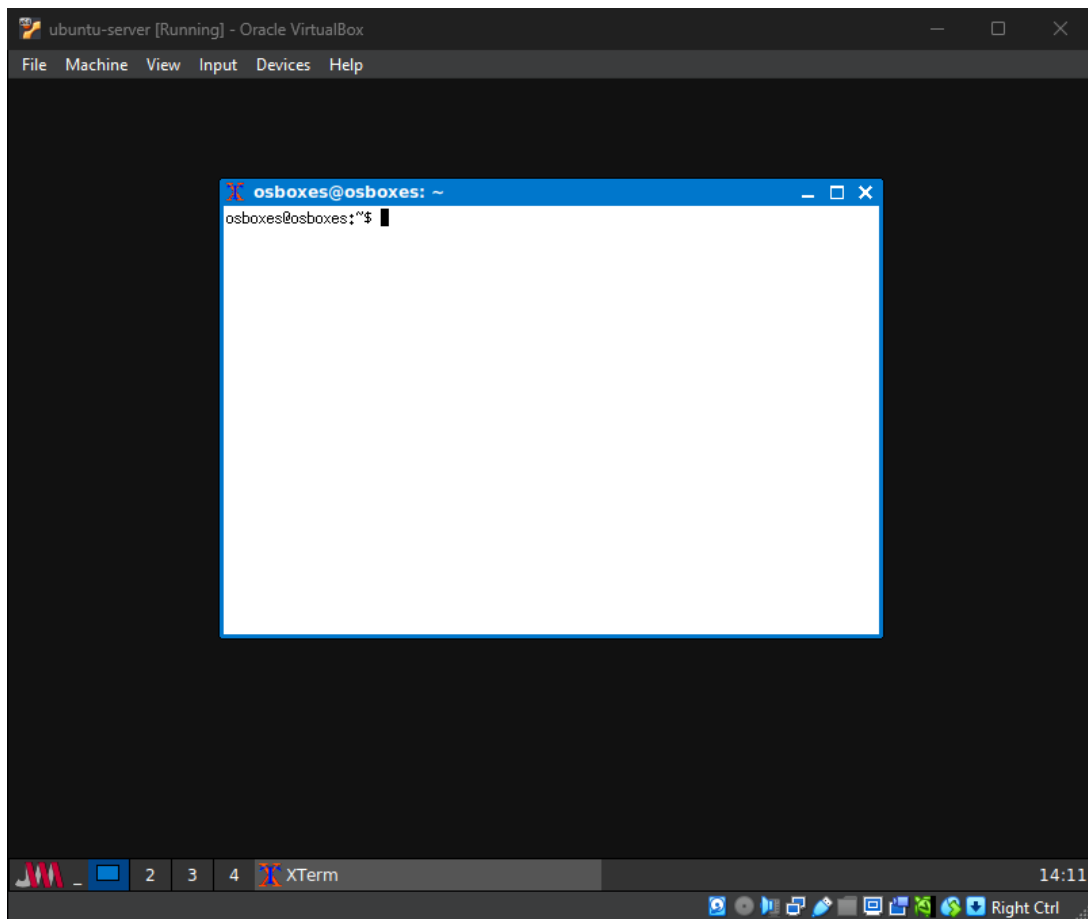
X.Org X Server 1.20.13
X Protocol Version 11, Revision 0
Build Operating System: linux Ubuntu
Current Operating System: Linux osboxes 5.4.0-100-generic #113-Ubuntu SMP Thu Feb 3 18:43:29 UTC 2022 x86_64
Kernel command line: BOOT_IMAGE=/vmlinuz-5.4.0-100-generic root=UUID=d850ca85-d161-402a-b05d-3e17a85a7c37 ro maybe-ubiquity
Build Date: 11 October 2024 02:23:55PM
xorg-server 2:1.20.13-1ubuntu1~20.04.18 (For technical support please see http://www.ubuntu.com/support)
Current version of pixman: 0.38.4
Before reporting problems, check http://wiki.x.org
to make sure that you have the latest version.
Markers: (--) probed, (**) from config file, (==) default setting,
(++) from command line, (!!) notice, (II) informational,
(WW) warning, (EE) error, (NI) not implemented, (??) unknown.
(==) Log file: "/home/osboxes/.local/share/xorg/Xorg.0.log", Time: Wed Nov 20 14:10:16 2024
(==) Using system config directory "/usr/share/X11/xorg.conf.d"
waiting for X server to shut down (II) Server terminated successfully (0). Closing log file. (0, Success)
osboxes@osboxes:~$ _
```

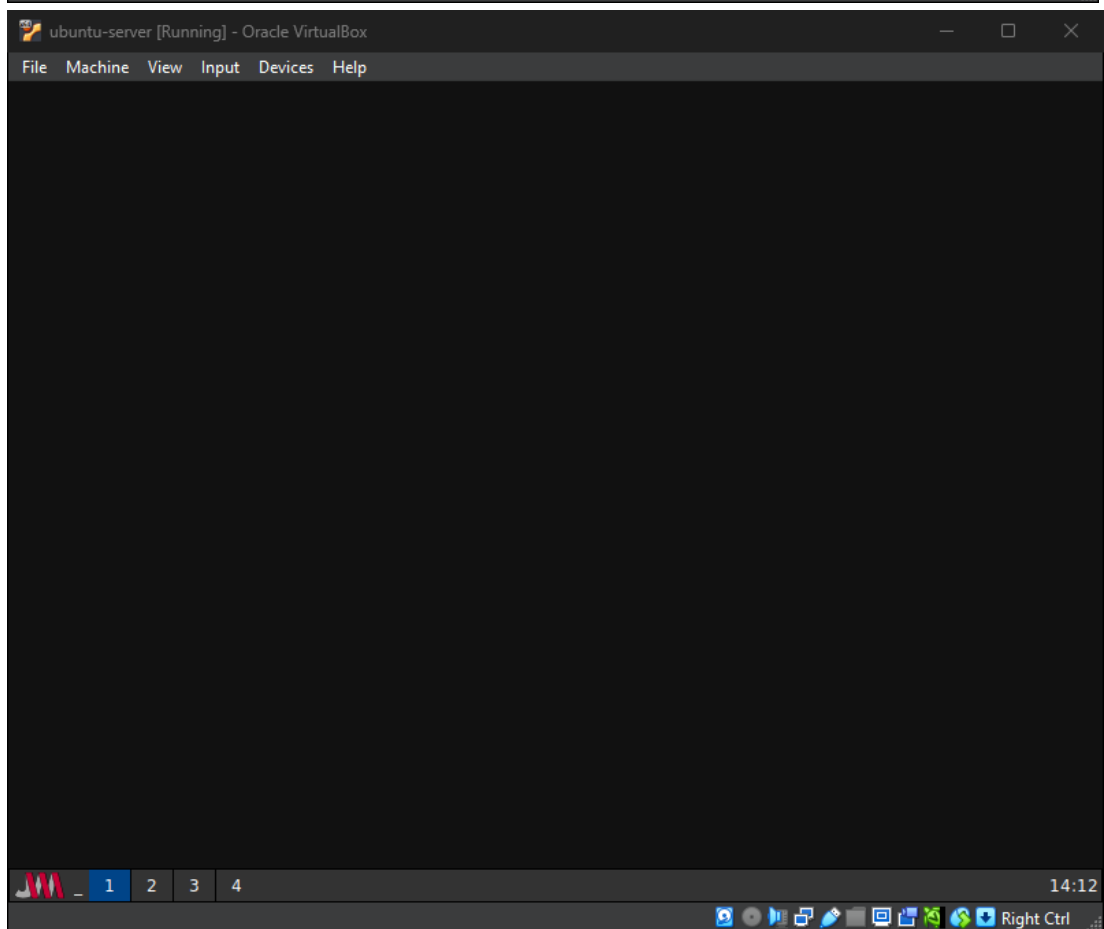
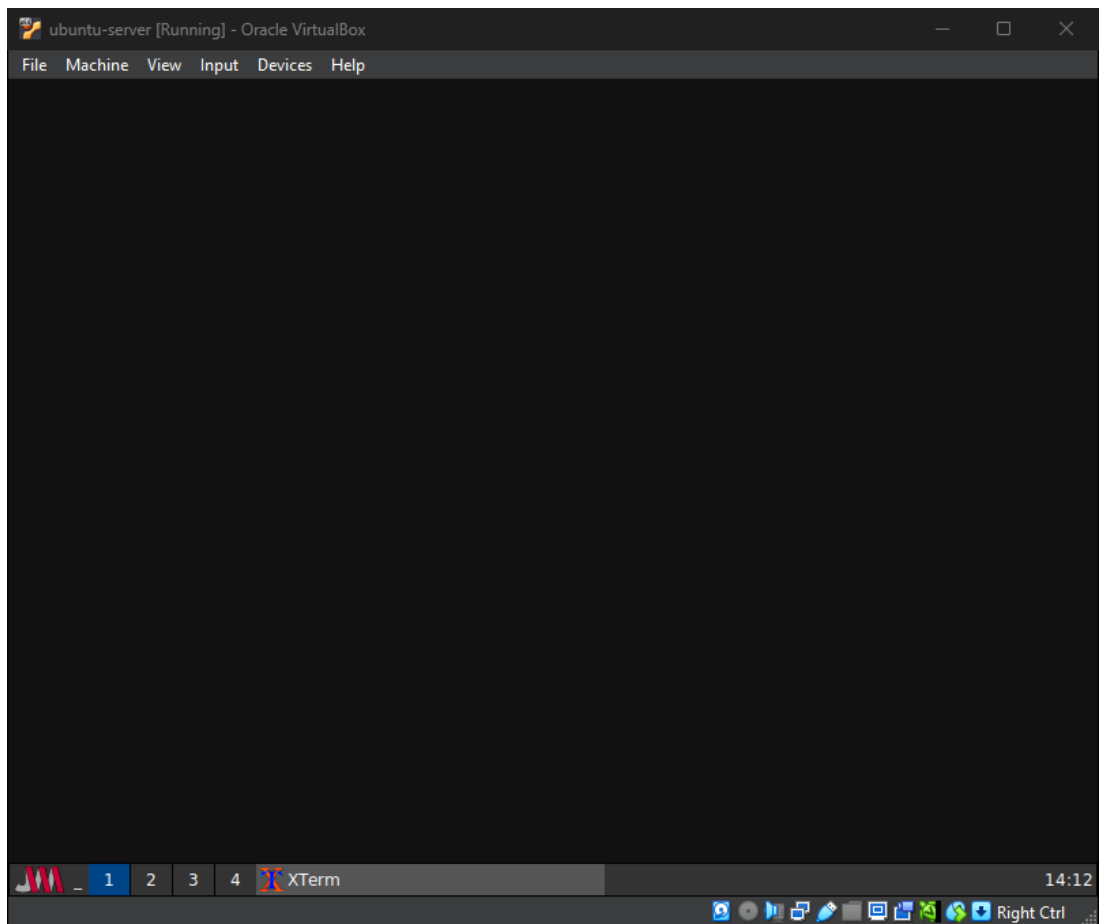
```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Setting up xserver-xorg-video-qxl (0.1.5+git20200331-1) ...
Setting up xserver-xorg-video-amdgpu (19.1.0-1ubuntu0.1) ...
Setting up xserver-xorg (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-video-nouveau (1:1.0.16-1) ...
Setting up xserver-xorg-input-libinput (0.29.0-1) ...
Setting up xserver-xorg-video-ati (1:19.1.0-1) ...
Setting up xserver-xorg-video-all (1:7.7+19ubuntu14) ...
Setting up xserver-xorg-input-all (1:7.7+19ubuntu14) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
osboxes@osboxes:~$ startx
xauth: file /home/osboxes/.Xauthority does not exist

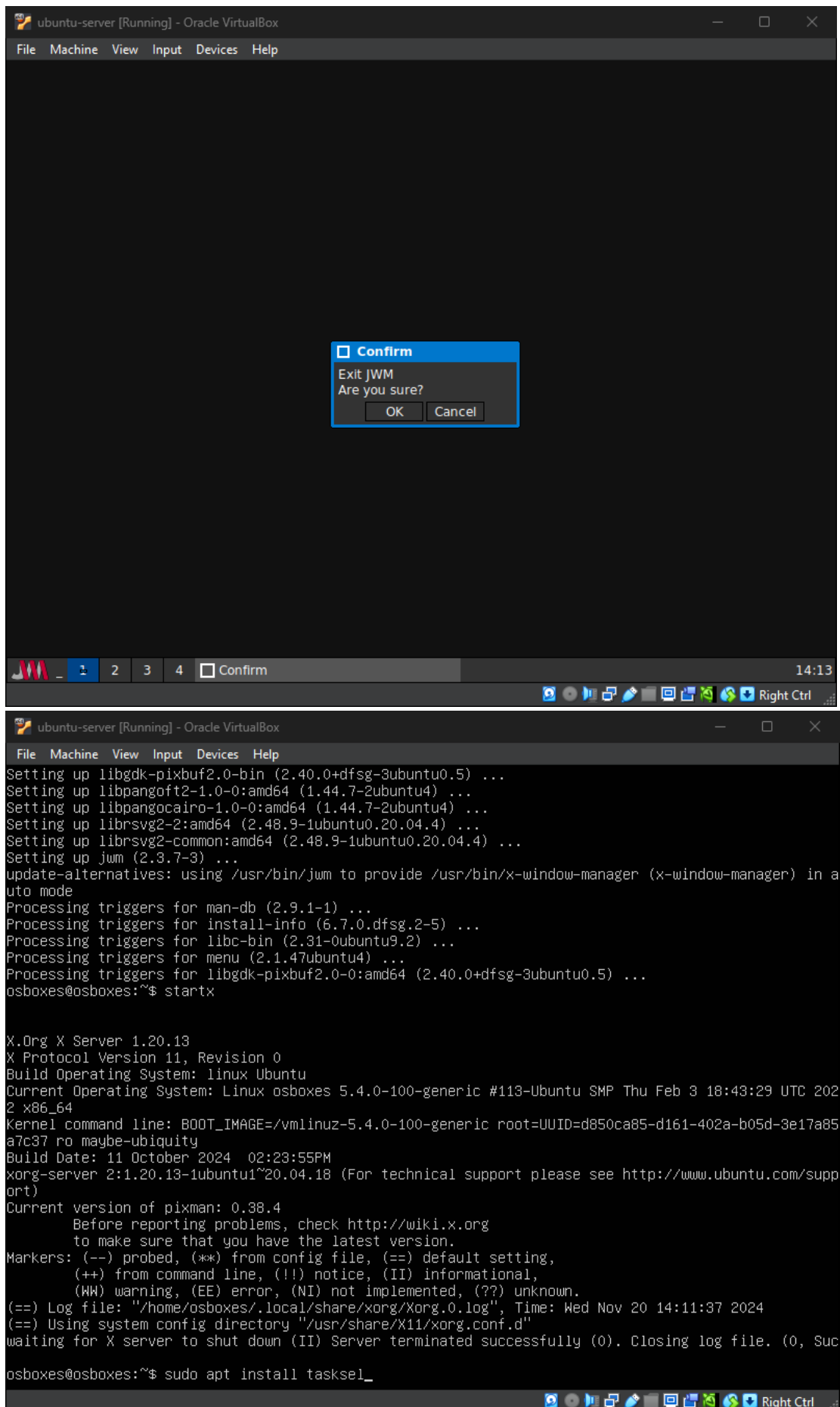
X.Org X Server 1.20.13
X Protocol Version 11, Revision 0
Build Operating System: linux Ubuntu
Current Operating System: Linux osboxes 5.4.0-100-generic #113-Ubuntu SMP Thu Feb 3 18:43:29 UTC 2022 x86_64
Kernel command line: BOOT_IMAGE=/vmlinuz-5.4.0-100-generic root=UUID=d850ca85-d161-402a-b05d-3e17a85a7c37 ro maybe-ubiquity
Build Date: 11 October 2024 02:23:55PM
xorg-server 2:1.20.13-1ubuntu1~20.04.18 (For technical support please see http://www.ubuntu.com/support)
Current version of pixman: 0.38.4
Before reporting problems, check http://wiki.x.org
to make sure that you have the latest version.
Markers: (--) probed, (**) from config file, (==) default setting,
(++) from command line, (!!) notice, (II) informational,
(WW) warning, (EE) error, (NI) not implemented, (??) unknown.
(==) Log file: "/home/osboxes/.local/share/xorg/Xorg.0.log", Time: Wed Nov 20 14:10:16 2024
(==) Using system config directory "/usr/share/X11/xorg.conf.d"
waiting for X server to shut down (II) Server terminated successfully (0). Closing log file. (0, Success)
osboxes@osboxes:~$ sudo apt install jwm
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Preparing to unpack .../06-libjbigo_2.1-3.1ubuntu0.20.04.1_amd64.deb ...
Unpacking libjbigo:amd64 (2.1-3.1ubuntu0.20.04.1) ...
Selecting previously unselected package libwebp6:amd64.
Preparing to unpack .../07-libwebp6_0.6.1-2ubuntu0.20.04.3_amd64.deb ...
Unpacking libwebp6:amd64 (0.6.1-2ubuntu0.20.04.3) ...
Selecting previously unselected package libtiff5:amd64.
Preparing to unpack .../08-libtiff5_4.1.0+git191117-2ubuntu0.20.04.14_amd64.deb ...
Unpacking libtiff5:amd64 (4.1.0+git191117-2ubuntu0.20.04.14) ...
Selecting previously unselected package libgdk-pixbuf2.0-common.
Preparing to unpack .../09-libgdk-pixbuf2.0-common_2.40.0+dfsg-3ubuntu0.5_all.deb ...
Unpacking libgdk-pixbuf2.0-common (2.40.0+dfsg-3ubuntu0.5) ...
Selecting previously unselected package libgdk-pixbuf2.0-0:amd64.
Preparing to unpack .../10-libgdk-pixbuf2.0-0_2.40.0+dfsg-3ubuntu0.5_amd64.deb ...
Unpacking libgdk-pixbuf2.0-0:amd64 (2.40.0+dfsg-3ubuntu0.5) ...
Selecting previously unselected package libgraphite2-3:amd64.
Preparing to unpack .../11-libgraphite2-3_1.3.13-11build1_amd64.deb ...
Unpacking libgraphite2-3:amd64 (1.3.13-11build1) ...
Selecting previously unselected package libharfbuzz0b:amd64.
Preparing to unpack .../12-libharfbuzz0b_2.6.4-1ubuntu4.2_amd64.deb ...
Unpacking libharfbuzz0b:amd64 (2.6.4-1ubuntu4.2) ...
Selecting previously unselected package libthai-data.
Preparing to unpack .../13-libthai-data_0.1.28-3_all.deb ...
Unpacking libthai-data (0.1.28-3) ...
Selecting previously unselected package libdatrie1:amd64.
Preparing to unpack .../14-libdatrie1_0.2.12-3_amd64.deb ...
Unpacking libdatrie1:amd64 (0.2.12-3) ...
Selecting previously unselected package libthai0:amd64.
Preparing to unpack .../15-libthai0_0.1.28-3_amd64.deb ...
Unpacking libthai0:amd64 (0.1.28-3) ...
Selecting previously unselected package libpango-1.0-0:amd64.
Preparing to unpack .../16-libpango-1.0-0_1.44.7-2ubuntu4_amd64.deb ...
Unpacking libpango-1.0-0:amd64 (1.44.7-2ubuntu4) ...
Selecting previously unselected package libpangoft2-1.0-0:amd64.
Preparing to unpack .../17-libpangoft2-1.0-0_1.44.7-2ubuntu4_amd64.deb ...
Unpacking libpangoft2-1.0-0:amd64 (1.44.7-2ubuntu4) ...
Progress: [ 36%] [#####.....]
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Preparing to unpack .../23-menu_2.1.47ubuntu4_amd64.deb ...
Unpacking menu (2.1.47ubuntu4) ...
Setting up libgraphite2-3:amd64 (1.3.13-11build1) ...
Setting up fontconfig (2.13.1-2ubuntu3) ...
Regenerating fonts cache... done.
Setting up libdatrie1:amd64 (0.2.12-3) ...
Setting up libxcb-render0:amd64 (1.14-2) ...
Setting up libgdk-pixbuf2.0-common (2.40.0+dfsg-3ubuntu0.5) ...
Setting up libjbigo:amd64 (2.1-3.1ubuntu0.20.04.1) ...
Setting up libcairo2:amd64 (1.16.0-4ubuntu1) ...
Setting up libwebp6:amd64 (0.6.1-2ubuntu0.20.04.3) ...
Setting up libjpeg-turbo8:amd64 (2.0.3-0ubuntu1.20.04.3) ...
Setting up libharfbuzz0b:amd64 (2.6.4-1ubuntu4.2) ...
Setting up libthai-data (0.1.28-3) ...
Setting up libcairo-gobject2:amd64 (1.16.0-4ubuntu1) ...
Setting up menu (2.1.47ubuntu4) ...
Setting up libjpeg8:amd64 (8c-2ubuntu8) ...
Setting up libthai0:amd64 (0.1.28-3) ...
Setting up libtiff5:amd64 (4.1.0+git191117-2ubuntu0.20.04.14) ...
Setting up libpango-1.0-0:amd64 (1.44.7-2ubuntu4) ...
Setting up libgdk-pixbuf2.0-0:amd64 (2.40.0+dfsg-3ubuntu0.5) ...
Setting up libgdk-pixbuf2.0-bin (2.40.0+dfsg-3ubuntu0.5) ...
Setting up libpangoft2-1.0-0:amd64 (1.44.7-2ubuntu4) ...
Setting up libpangocairo-1.0-0:amd64 (1.44.7-2ubuntu4) ...
Setting up librsvg2-2:amd64 (2.48.9-1ubuntu0.20.04.4) ...
Setting up librsvg2-common:amd64 (2.48.9-1ubuntu0.20.04.4) ...
Setting up jwm (2.3.7-3) ...
update-alternatives: using /usr/bin/jwm to provide /usr/bin/x-window-manager (x-window-manager) in a
uto mode
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
Processing triggers for menu (2.1.47ubuntu4) ...
Processing triggers for libgdk-pixbuf2.0-0:amd64 (2.40.0+dfsg-3ubuntu0.5) ...
osboxes@osboxes:~$ _
```

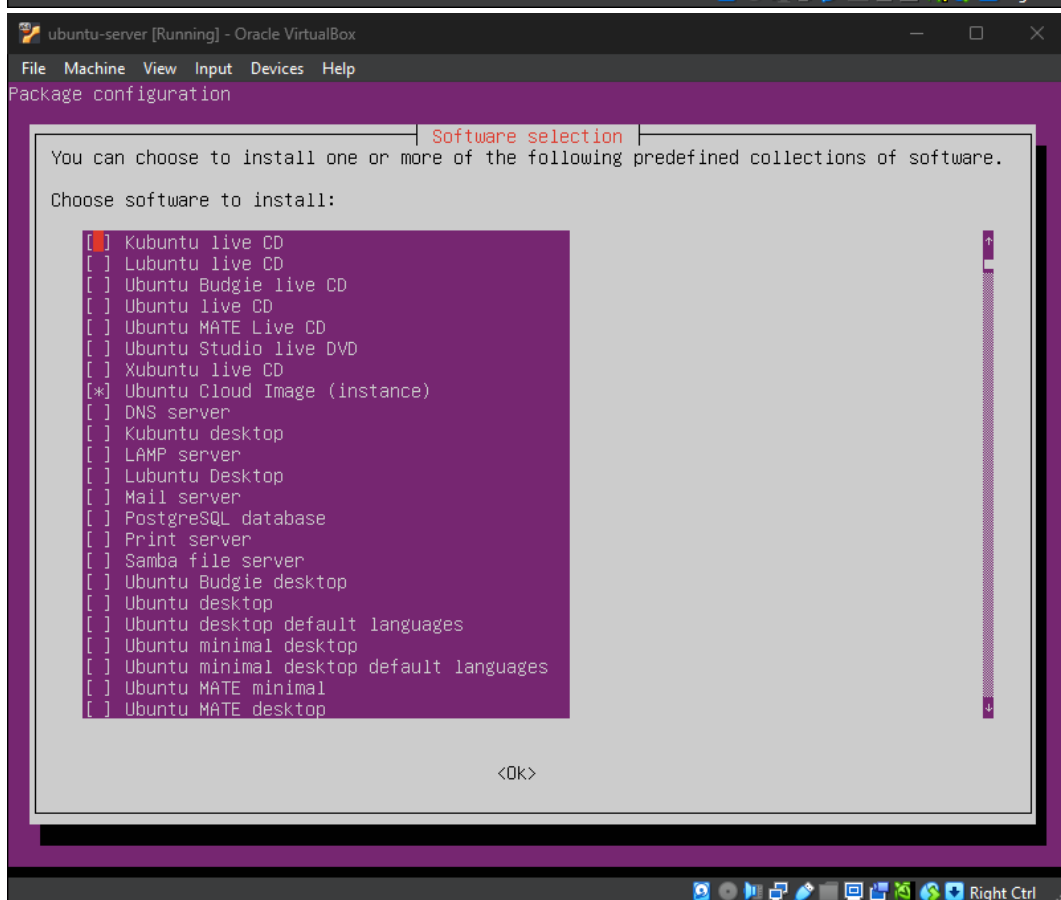


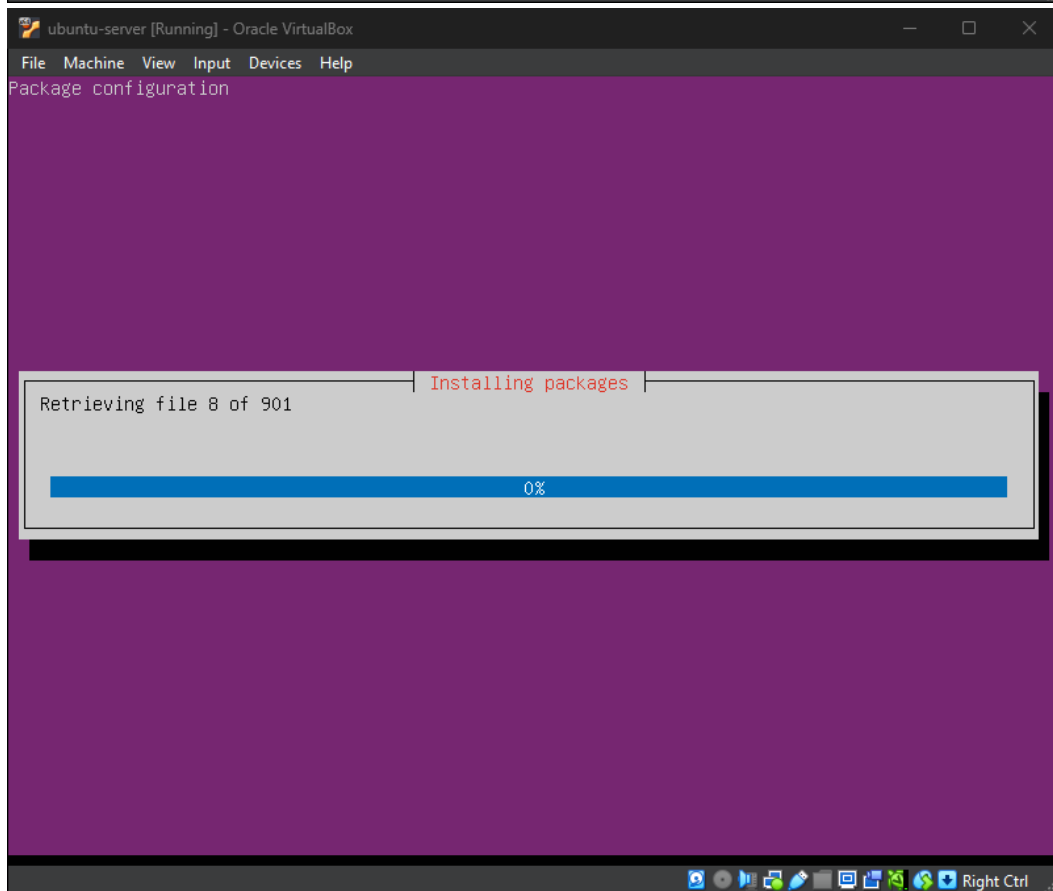
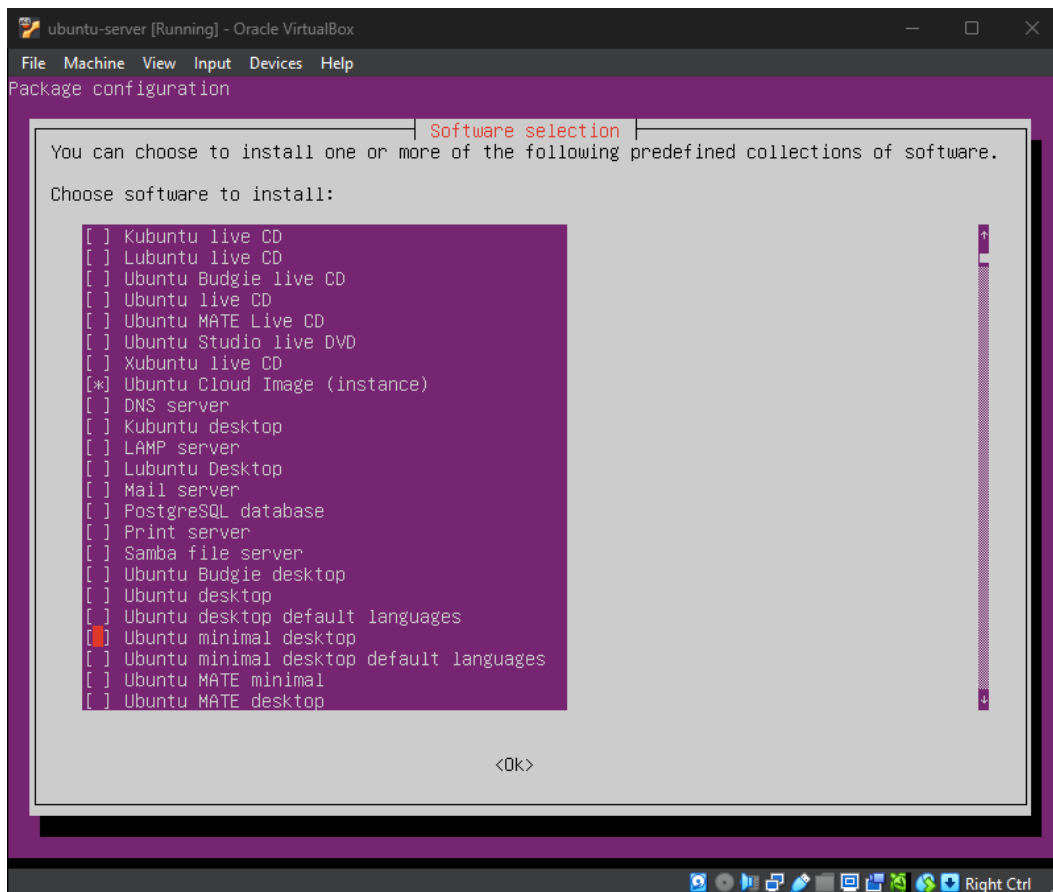




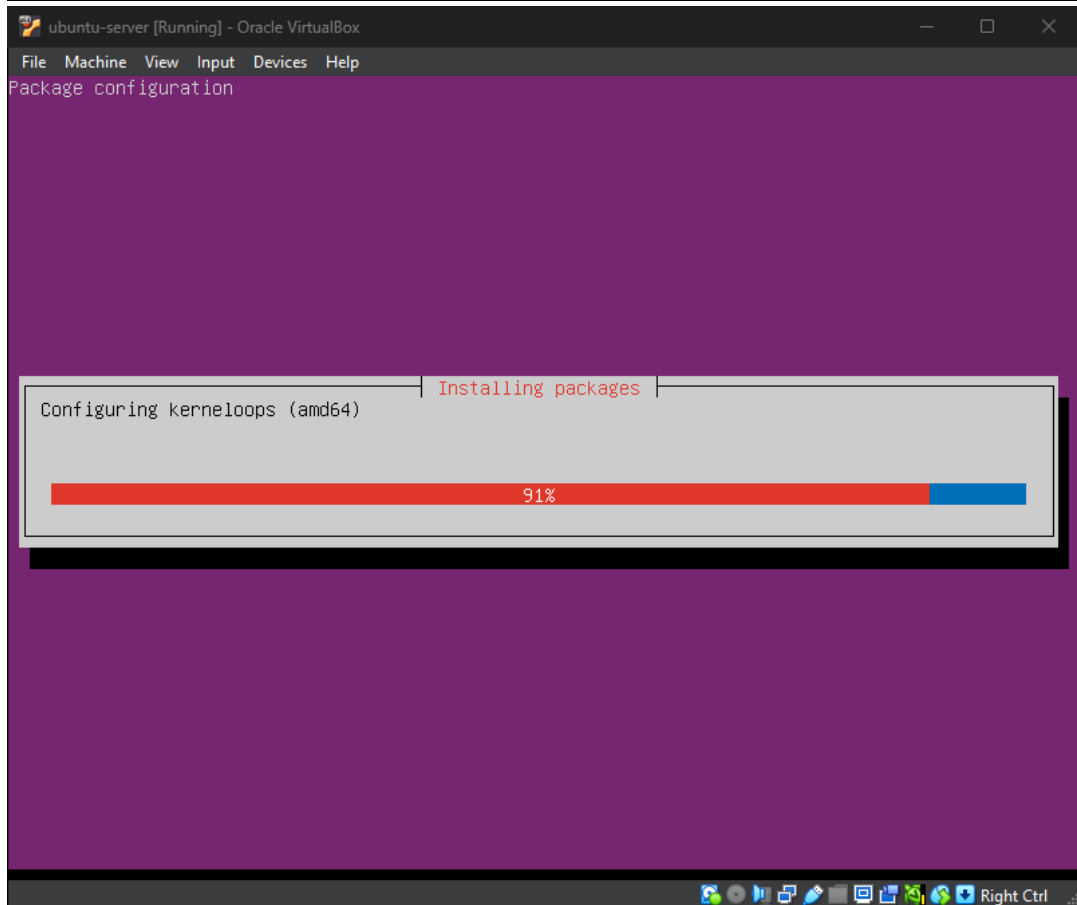
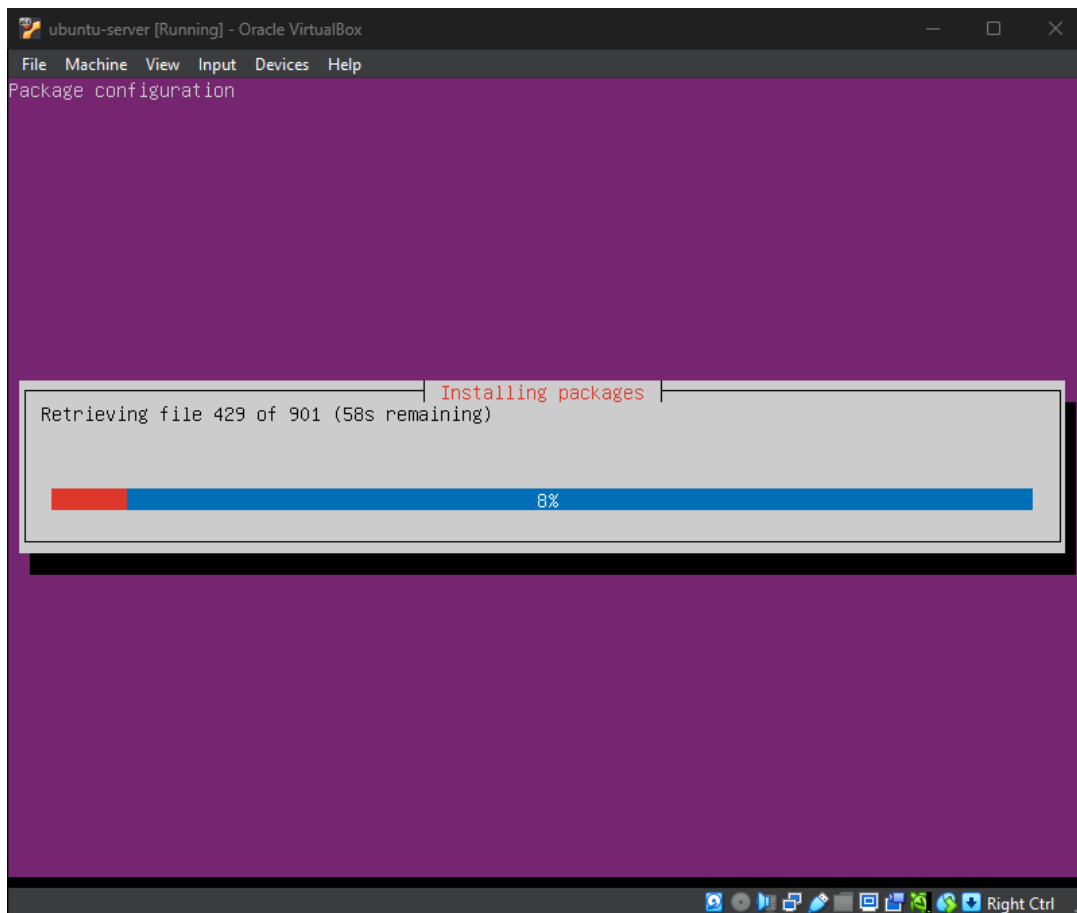
```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
waiting for X server to shut down (II) Server terminated successfully (0). Closing log file. (0, Suc

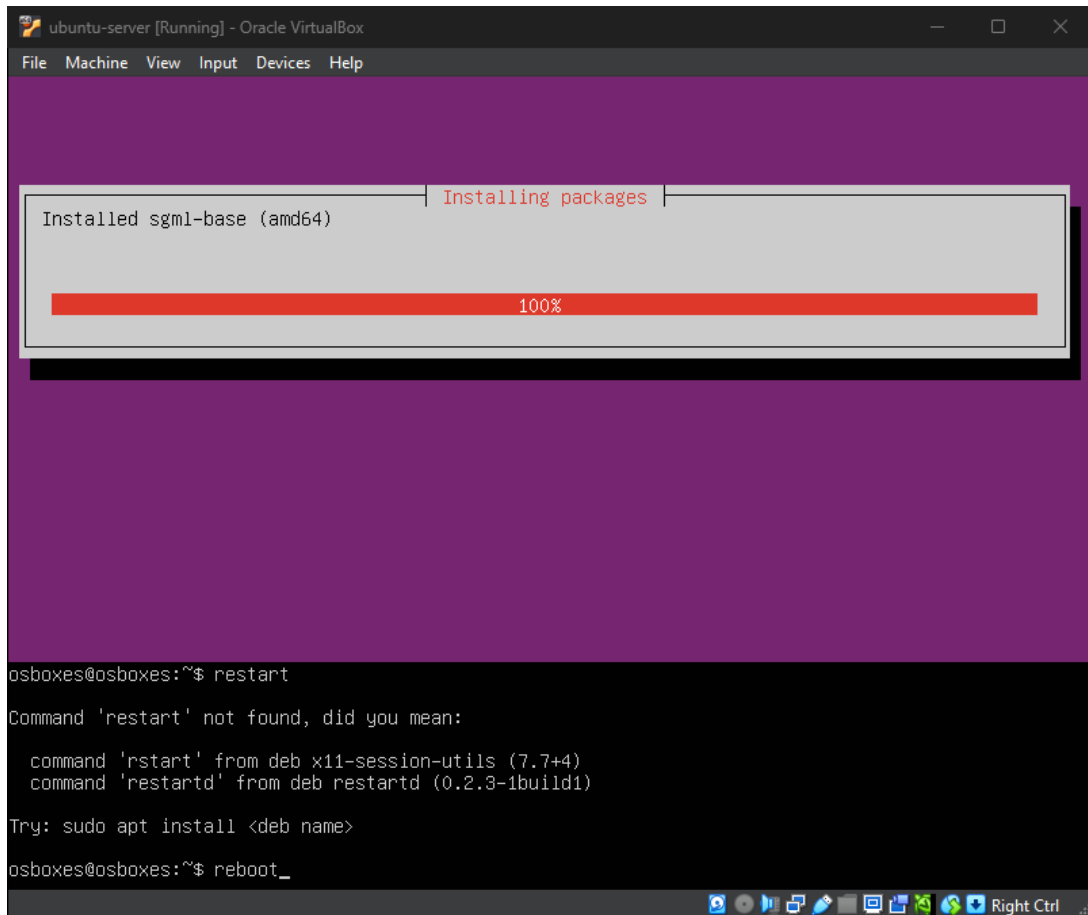
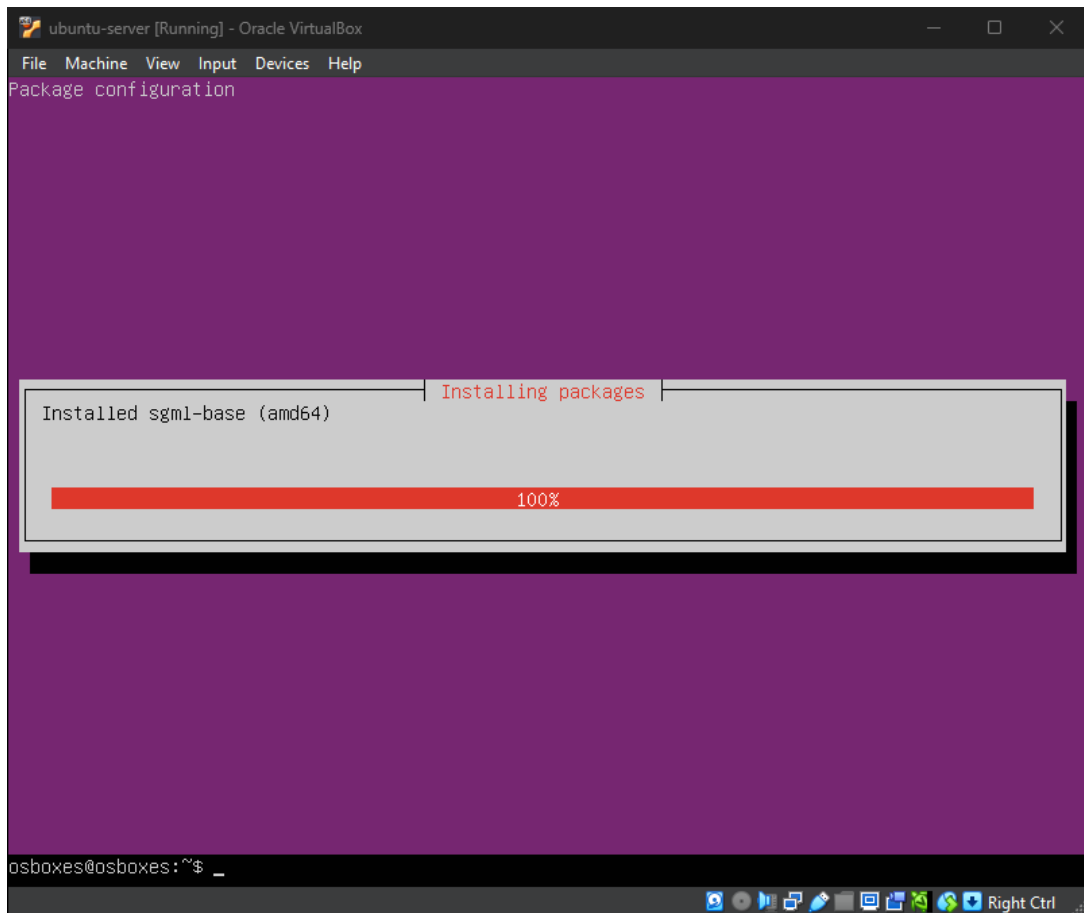
osboxes@osboxes:~$ sudo apt install tasksel
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  laptop-detect tasksel-data
The following NEW packages will be installed:
  laptop-detect tasksel tasksel-data
0 upgraded, 3 newly installed, 0 to remove and 246 not upgraded.
Need to get 40.0 kB of archives.
After this operation, 309 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu focal/main amd64 tasksel-data all 3.34ubuntu16 [5,340 B]
Get:2 http://archive.ubuntu.com/ubuntu focal/main amd64 tasksel all 3.34ubuntu16 [28.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal/main amd64 laptop-detect all 0.16 [6,016 B]
Fetched 40.0 kB in 1s (47.3 kB/s)
Preconfiguring packages ...
Selecting previously unselected package tasksel-data.
(Reading database ... 71234 files and directories currently installed.)
Preparing to unpack .../tasksel-data_3.34ubuntu16_all.deb ...
Unpacking tasksel-data (3.34ubuntu16) ...
Selecting previously unselected package tasksel.
Preparing to unpack .../tasksel_3.34ubuntu16_all.deb ...
Unpacking tasksel (3.34ubuntu16) ...
Selecting previously unselected package laptop-detect.
Preparing to unpack .../laptop-detect_0.16_all.deb ...
Unpacking laptop-detect (0.16) ...
Setting up laptop-detect (0.16) ...
Setting up tasksel (3.34ubuntu16) ...
Setting up tasksel-data (3.34ubuntu16) ...
Processing triggers for menu (2.1.47ubuntu4) ...
Processing triggers for man-db (2.9.1-1) ...
```



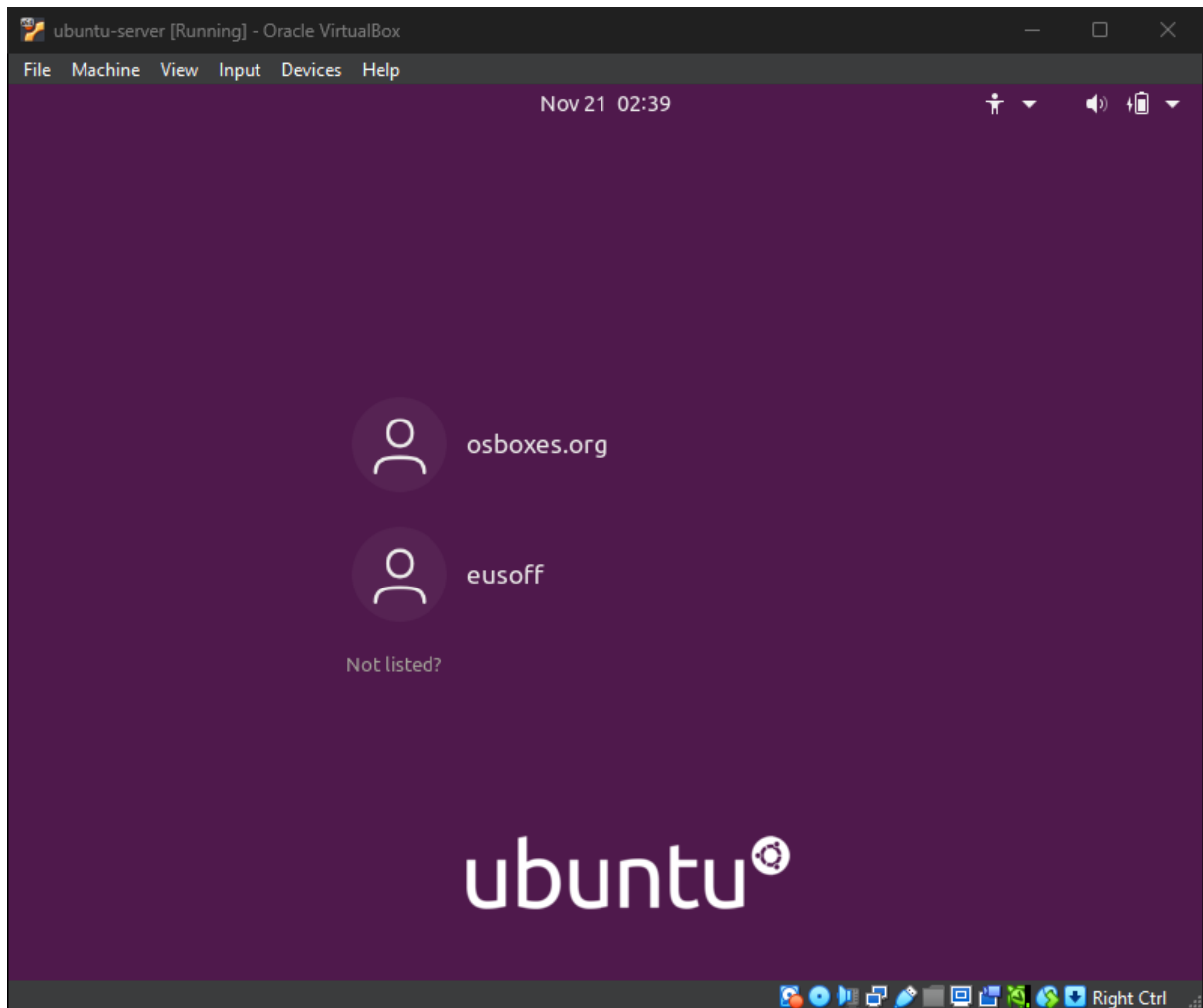


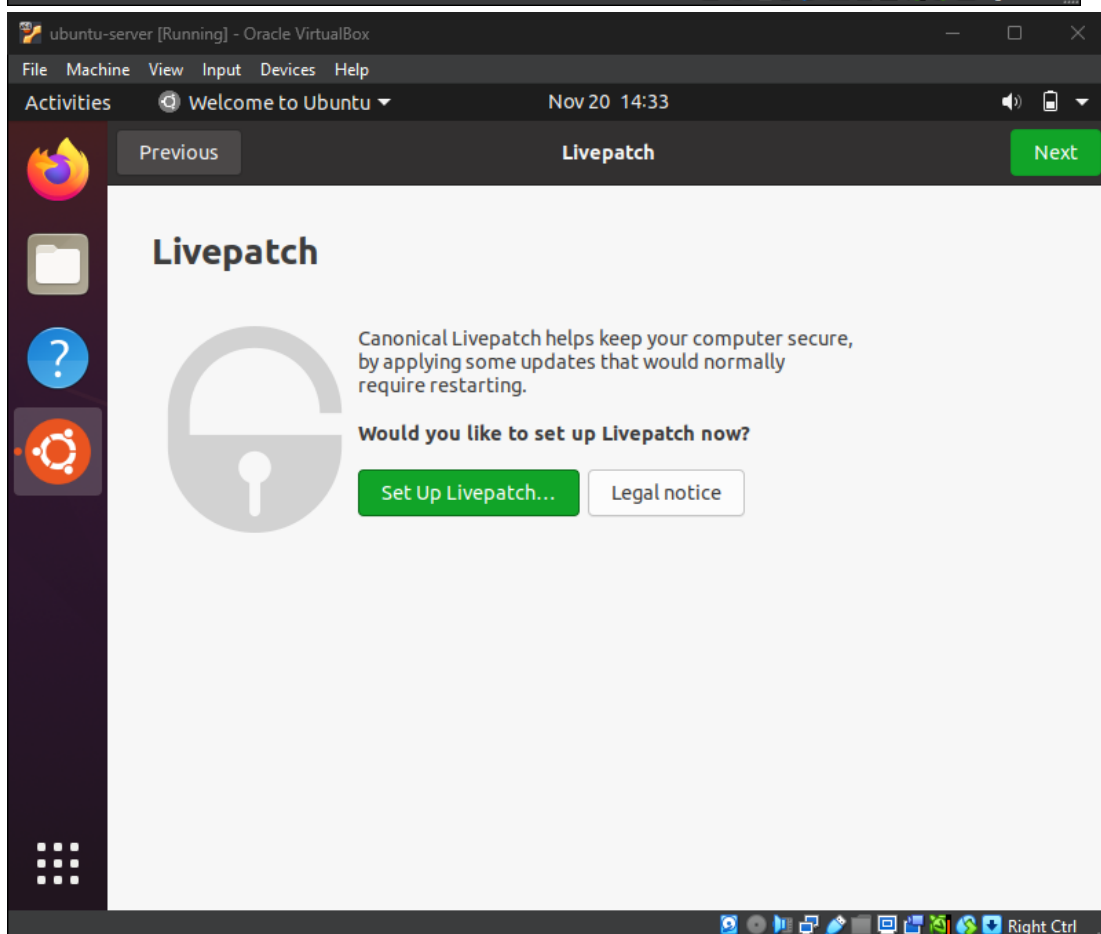
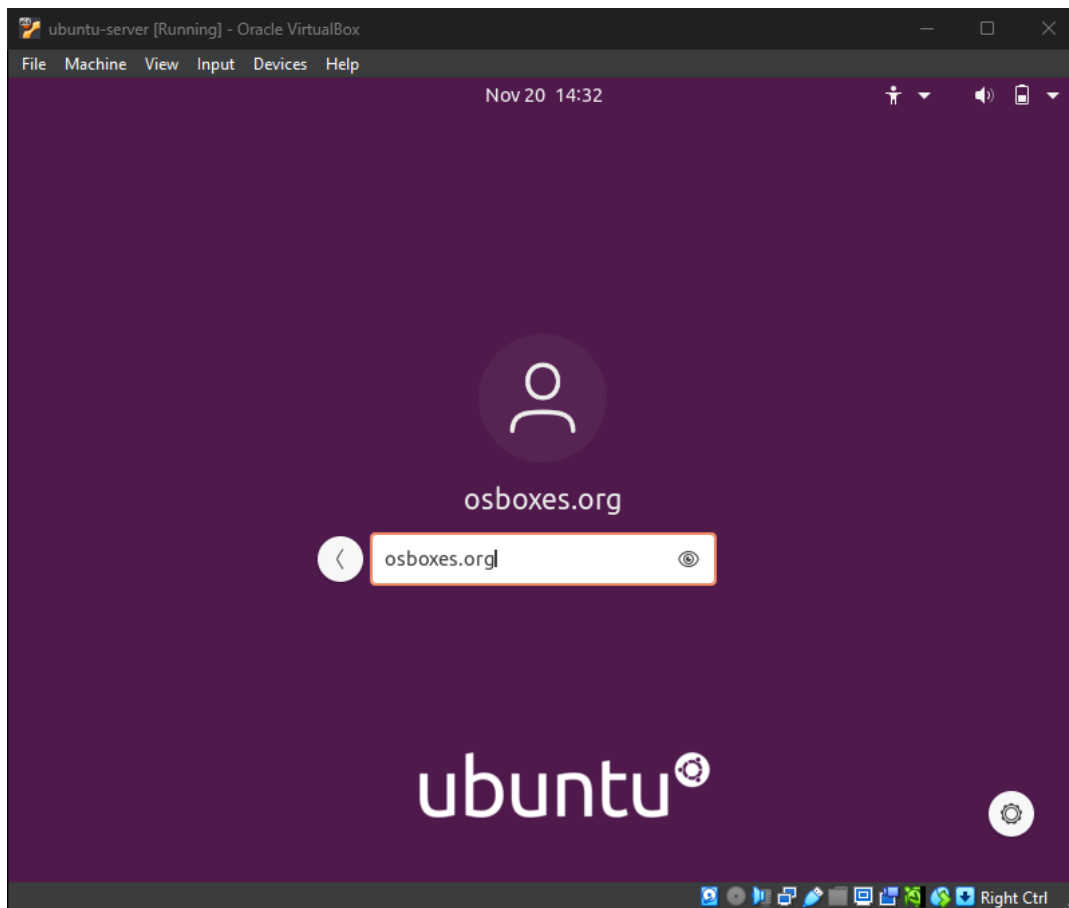
sudo tasksel if permission denied





```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[ 2.798873] rtc_cmos rtc_cmos: registered as rtc0
[ 2.803158] rtc_cmos rtc_cmos: alarms up to one day, 114 bytes nvram
[ 2.806647] i2c /dev entries driver
[ 2.809086] device-mapper: uevent: version 1.0.3
[ 2.813239] device-mapper: ioctl: 4.41.0-ioctl (2019-09-16) initialised: dm-d
evel@redhat.com
[ 2.818364] sr 1:0:0:0: [sr0] scsi3-mmc drive: 32x/32x xa/form2 tray
[ 2.818428] platform eisa.0: Probing EISA bus 0
[ 2.820647] cdrom: Uniform CD-ROM driver Revision: 3.20
[ 2.823790] platform eisa.0: EISA: Cannot allocate resource for mainboard
[ 2.829334] platform eisa.0: Cannot allocate resource for EISA slot 1
[ 2.831938] platform eisa.0: Cannot allocate resource for EISA slot 2
[ 2.834818] platform eisa.0: Cannot allocate resource for EISA slot 3
[ 2.837537] platform eisa.0: Cannot allocate resource for EISA slot 4
[ 2.840093] platform eisa.0: Cannot allocate resource for EISA slot 5
[ 2.843642] platform eisa.0: Cannot allocate resource for EISA slot 6
[ 2.846450] platform eisa.0: Cannot allocate resource for EISA slot 7
[ 2.850654] platform eisa.0: Cannot allocate resource for EISA slot 8
[ 2.856428] platform eisa.0: EISA: Detected 0 cards
[ 2.860576] intel_pstate: CPU model not supported
[ 2.867208] ledtrig-cpu: registered to indicate activity on CPUs
[ 2.871413] drop_monitor: Initializing network drop monitor service
[ 2.875799] sr 1:0:0:0: Attached scsi generic sg0 type 5
[ 2.877997] NET: Registered protocol family 10
```






```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:38

osboxes@osboxes: ~
Metadata [212 B]
Get:54 http://archive.ubuntu.com/ubuntu focal-security/restricted DEP-11 48x48
Icons [29 B]
Get:55 http://archive.ubuntu.com/ubuntu focal-security/restricted DEP-11 64x64
Icons [29 B]
Get:56 http://archive.ubuntu.com/ubuntu focal-security/restricted DEP-11 64x64@
2 Icons [29 B]
Get:57 http://archive.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Me
tadata [160 kB]
Get:58 http://archive.ubuntu.com/ubuntu focal-security/universe DEP-11 48x48 Ic
d Terminal [kB]
Get:59 http://archive.ubuntu.com/ubuntu focal-security/universe DEP-11 64x64 Ic
ons [166 kB]
Get:60 http://archive.ubuntu.com/ubuntu focal-security/universe DEP-11 64x64@2
Icons [29 B]
Get:61 http://archive.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11
Metadata [940 B]
Get:62 http://archive.ubuntu.com/ubuntu focal-security/multiverse DEP-11 48x48
Icons [1,867 B]
Get:63 http://archive.ubuntu.com/ubuntu focal-security/multiverse DEP-11 64x64
Icons [2,497 B]
Get:64 http://archive.ubuntu.com/ubuntu focal-security/multiverse DEP-11 64x64@
2 Icons [29 B]
Fetched 17.9 MB in 11s (1,592 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
osboxes@osboxes:~$
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:39

osboxes@osboxes: ~
Fetched 17.9 MB in 11s (1,592 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
osboxes@osboxes:~$ sudo apt install virtualbox-guest-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  virtualbox-guest-x11
The following NEW packages will be installed:
  virtualbox-guest-utils
0 upgraded, 1 newly installed, 0 to remove and 155 not upgraded.
Need to get 893 kB of archives.
After this operation, 6,087 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 virtualbo
x-guest-utils amd64 6.1.50-dfsg-1-ubuntu1.20.04.1 [893 kB]
Fetched 893 kB in 2s (387 kB/s)
Selecting previously unselected package virtualbox-guest-utils.
(Reading database ... 128098 files and directories currently installed.)
Preparing to unpack .../virtualbox-guest-utils_6.1.50-dfsg-1-ubuntu1.20.04.1_am
d64.deb ...
Unpacking virtualbox-guest-utils (6.1.50-dfsg-1-ubuntu1.20.04.1) ...
Setting up virtualbox-guest-utils (6.1.50-dfsg-1-ubuntu1.20.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/virtualbox-guest-ut
ils.service to /lib/systemd/system/virtualbox-guest-utils.service.
Show Applications
Progress: [ 60%] [#####]
```

ubuntu-server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Nov 20 14:39

osboxes@osboxes: ~

```
Reading state information... Done
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
osboxes@osboxes:~$ sudo apt install virtualbox-guest-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  virtualbox-guest-x11
The following NEW packages will be installed:
  virtualbox-guest-utils
0 upgraded, 1 newly installed, 0 to remove and 155 not upgraded.
Need to get 893 kB of archives.
After this operation, 6,087 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 virtualbo
x-guest-utils amd64 6.1.50-dfsg-1~ubuntu1.20.04.1 [893 kB]
Fetched 893 kB in 2s (387 kB/s)
Selecting previously unselected package virtualbox-guest-utils.
(Reading database ... 128098 files and directories currently installed.)
Preparing to unpack .../virtualbox-guest-utils_6.1.50-dfsg-1~ubuntu1.20.04.1_am
d64.deb ...
Unpacking virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Setting up virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/virtualbox-guest-ut
ils.service → /lib/systemd/system/virtualbox-guest-utils.service.
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
osboxes@osboxes:~$
```

ubuntu-server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Nov 20 14:40

osboxes@osboxes: ~

```
d64.deb ...
Unpacking virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Setting up virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/virtualbox-guest-ut
ils.service → /lib/systemd/system/virtualbox-guest-utils.service.
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
osboxes@osboxes:~$ sudo apt install virtualbox-guest-x11
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  virtualbox-guest-x11
0 upgraded, 1 newly installed, 0 to remove and 155 not upgraded.
Need to get 566 kB of archives.
After this operation, 2,133 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 virtualbo
x-guest-x11 amd64 6.1.50-dfsg-1~ubuntu1.20.04.1 [566 kB]
Fetched 566 kB in 2s (269 kB/s)
Selecting previously unselected package virtualbox-guest-x11.
(Reading database ... 128112 files and directories currently installed.)
Preparing to unpack .../virtualbox-guest-x11_6.1.50-dfsg-1~ubuntu1.20.04.1_amd6
4.deb ...
Unpacking virtualbox-guest-x11 (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Setting up virtualbox-guest-x11 (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Processing triggers for man-db (2.9.1-1) ...
Show Applications
```

ubuntu-server [Running] - Oracle VirtualBox

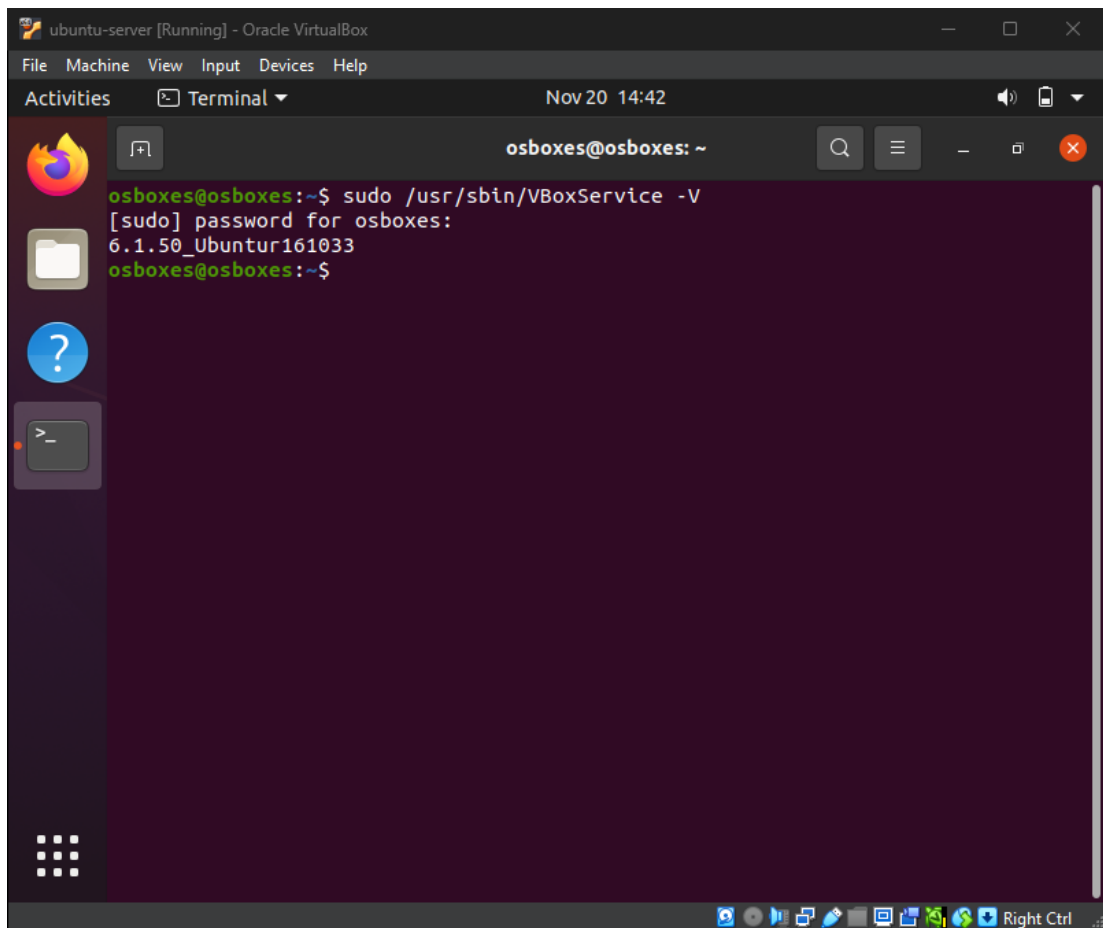
File Machine View Input Devices Help

Activities Terminal Nov 20 14:40

osboxes@osboxes: ~

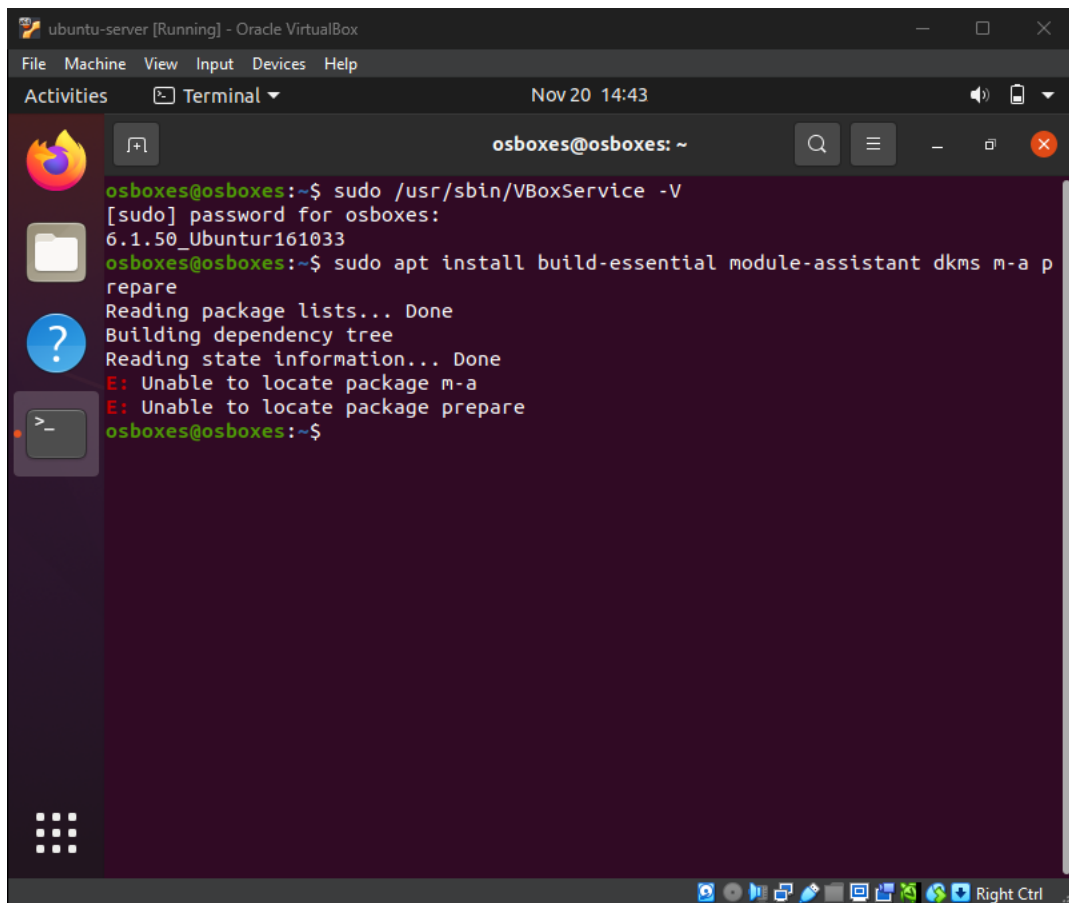
```
d64.deb ...
Unpacking virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Setting up virtualbox-guest-utils (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/virtualbox-guest-ut
ils.service → /lib/systemd/system/virtualbox-guest-utils.service.
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
osboxes@osboxes:~$ sudo apt install virtualbox-guest-x11
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  virtualbox-guest-x11
0 upgraded, 1 newly installed, 0 to remove and 155 not upgraded.
Need to get 566 kB of archives.
After this operation, 2,133 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 virtualbo
x-guest-x11 amd64 6.1.50-dfsg-1~ubuntu1.20.04.1 [566 kB]
Fetched 566 kB in 2s (269 kB/s)
Selecting previously unselected package virtualbox-guest-x11.
(Reading database ... 128112 files and directories currently installed.)
Preparing to unpack .../virtualbox-guest-x11_6.1.50-dfsg-1~ubuntu1.20.04.1_amd6
4.deb ...
Unpacking virtualbox-guest-x11 (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Setting up virtualbox-guest-x11 (6.1.50-dfsg-1~ubuntu1.20.04.1) ...
Processing triggers for man-db (2.9.1-1) ...
osboxes@osboxes:~$ sudo reboot
```

Right Ctrl



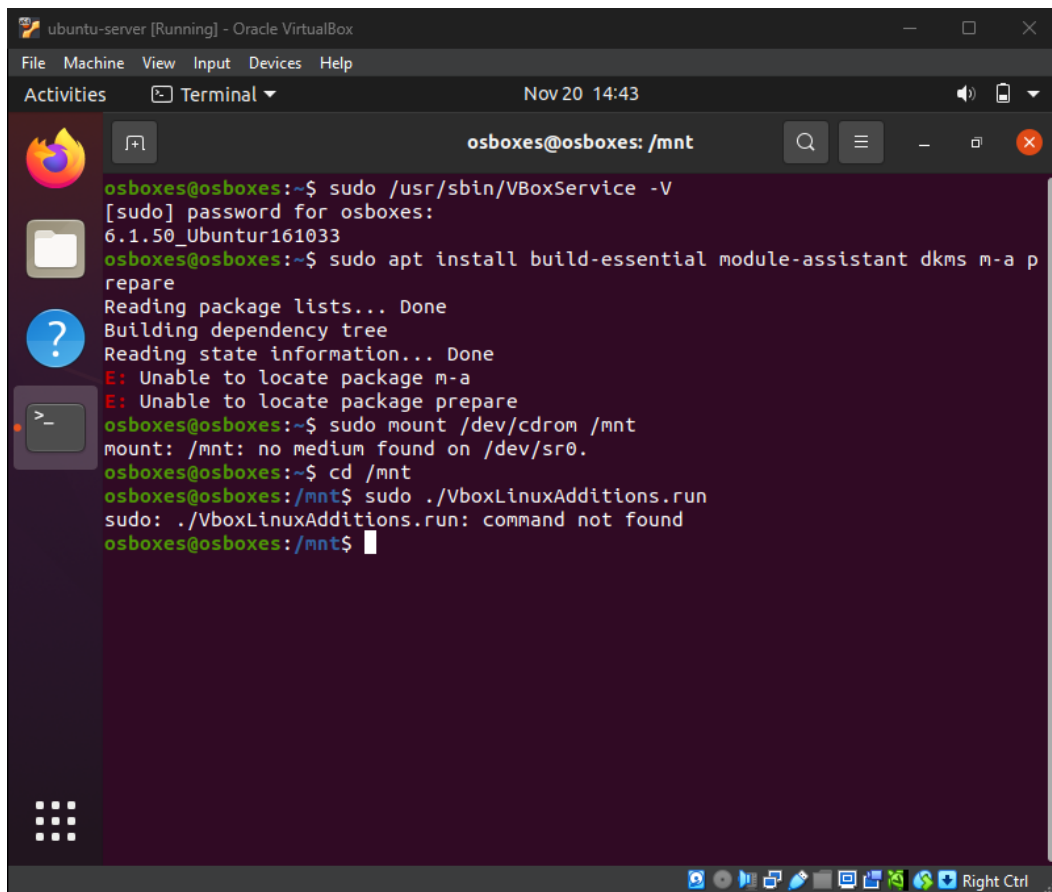
```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:42
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo /usr/sbin/VBoxService -V
[sudo] password for osboxes:
6.1.50_Ubuntur161033
osboxes@osboxes:~$
```

no?



```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:43
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo /usr/sbin/VBoxService -V
[sudo] password for osboxes:
6.1.50_Ubuntur161033
osboxes@osboxes:~$ sudo apt install build-essential module-assistant dkms m-a p
repare
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package m-a
E: Unable to locate package prepare
osboxes@osboxes:~$
```

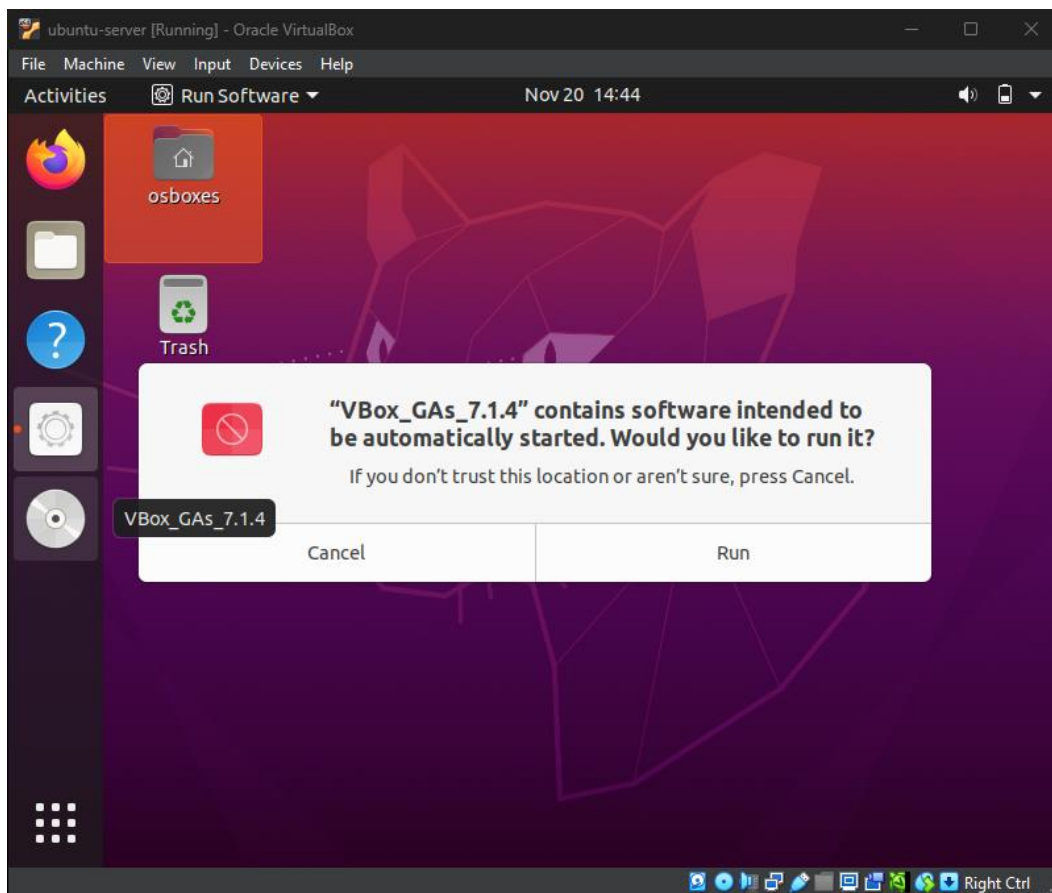
no?

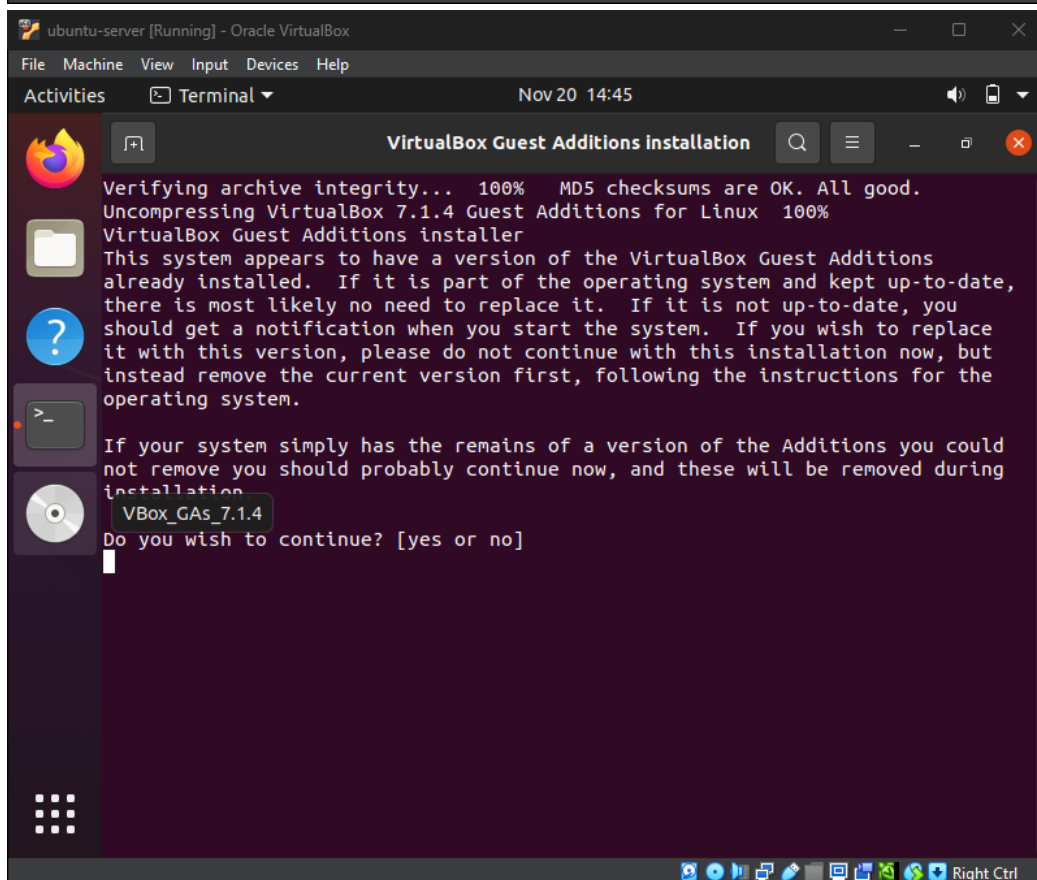
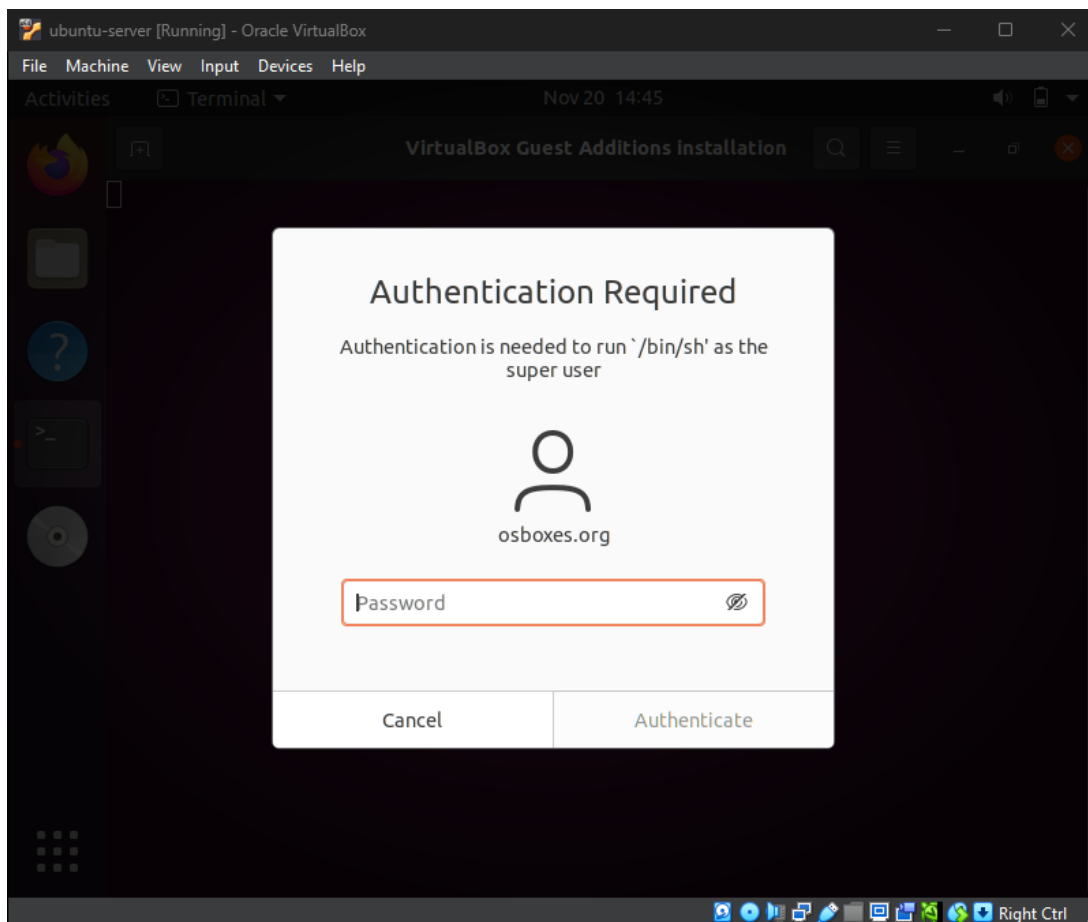


The image shows a terminal window titled 'ubuntu-server [Running] - Oracle VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with 'Activities' and a 'Terminal' dropdown. The terminal shows the following commands and output:

```
osboxes@osboxes:~$ sudo /usr/sbin/VBoxService -V
[sudo] password for osboxes:
6.1.50_Ubuntu161033
osboxes@osboxes:~$ sudo apt install build-essential module-assistant dkms m-a p
repare
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package m-a
E: Unable to locate package prepare
osboxes@osboxes:~$ sudo mount /dev/cdrom /mnt
mount: /mnt: no medium found on /dev/sr0.
osboxes@osboxes:~$ cd /mnt
osboxes@osboxes:/mnt$ sudo ./VboxLinuxAdditions.run
sudo: ./VboxLinuxAdditions.run: command not found
osboxes@osboxes:/mnt$
```

no?





ubuntu-server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Nov 20 14:45

VirtualBox Guest Additions Installation

```
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.1.4 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
This system appears to have a version of the VirtualBox Guest Additions
already installed. If it is part of the operating system and kept up-to-date,
there is most likely no need to replace it. If it is not up-to-date, you
should get a notification when you start the system. If you wish to replace
it with this version, please do not continue with this installation now, but
instead remove the current version first, following the instructions for the
operating system.

If your system simply has the remains of a version of the Additions you could
not remove you should probably continue now, and these will be removed during
installation.

Do you wish to continue? [yes or no]
y
touch: cannot touch '/var/lib/VBoxGuestAdditions/skip-5.4.0-100-generic': No su
ch file or directory
```

no

ubuntu-server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Nov 20 14:45

VirtualBox Guest Additions Installation

```
If your system simply has the remains of a version of the Additions you could
not remove you should probably continue now, and these will be removed during
installation.

Do you wish to continue? [yes or no]
y
touch: cannot touch '/var/lib/VBoxGuestAdditions/skip-5.4.0-100-generic': No su
ch file or directory
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel 5.4.0-100-generic.

This system is currently not set up to build kernel modules.
Please install the gcc make perl packages from your distribution.
VirtualBox Guest Additions: Running kernel modules will not be replaced until
the system is restarted or 'rcvboxadd reload' triggered
VirtualBox Guest Additions: reloading kernel modules and services
VirtualBox Guest Additions: cannot reload kernel modules: one or more module(s)
is still in use
VirtualBox Guest Additions: kernel modules and services were not reloaded
The log file /var/log/vboxadd-setup.log may contain further information.
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:46

osboxes@osboxes: /mnt

osboxes@osboxes:~$ sudo apt install build-essential module-assistant dkms m-a p
repare
[Files] password for osboxes:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package m-a
E: Unable to locate package prepare
osboxes@osboxes:~$ sudo mount /dev/cdrom /mnt
mount: /mnt: WARNING: device write-protected, mounted read-only.
osboxes@osboxes:~$ cd /mnt
osboxes@osboxes:/mnt$ sudo ./VBoxLinuxAdditions.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.1.4 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
Removing installed version 7.1.4 of VirtualBox Guest Additions...
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel 5.4.0-100-generic.

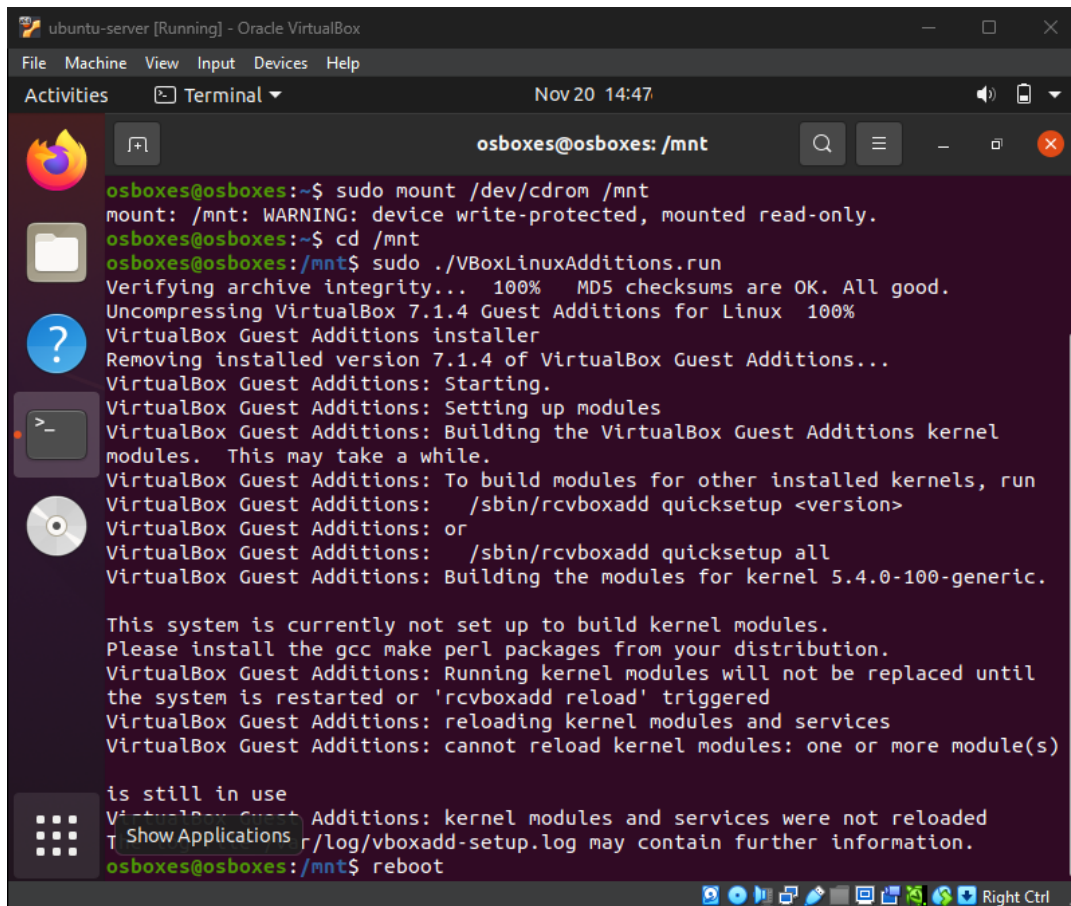
This system is currently not set up to build kernel modules.
Please install the gcc make perl packages from your distribution.
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 20 14:47

osboxes@osboxes: /mnt

osboxes@osboxes:~$ sudo mount /dev/cdrom /mnt
mount: /mnt: WARNING: device write-protected, mounted read-only.
osboxes@osboxes:~$ cd /mnt
osboxes@osboxes:/mnt$ sudo ./VBoxLinuxAdditions.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.1.4 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
Removing installed version 7.1.4 of VirtualBox Guest Additions...
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel 5.4.0-100-generic.

This system is currently not set up to build kernel modules.
Please install the gcc make perl packages from your distribution.
VirtualBox Guest Additions: Running kernel modules will not be replaced until
the system is restarted or 'rcvboxadd reload' triggered
VirtualBox Guest Additions: reloading kernel modules and services
VirtualBox Guest Additions: cannot reload kernel modules: one or more module(s)
is still in use
VirtualBox Guest Additions: kernel modules and services were not reloaded
The log file /var/log/vboxadd-setup.log may contain further information.
osboxes@osboxes:/mnt$
```



The screenshot shows a terminal window titled 'ubuntu-server [Running] - Oracle VM VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with 'Activities' and a 'Terminal' dropdown. The date and time 'Nov 20 14:47' are displayed. The terminal window itself has a title bar 'osboxes@osboxes: /mnt' and a search icon. The terminal output shows the following commands and their results:

```
osboxes@osboxes:~$ sudo mount /dev/cdrom /mnt
mount: /mnt: WARNING: device write-protected, mounted read-only.
osboxes@osboxes:~$ cd /mnt
osboxes@osboxes:/mnt$ sudo ./VBoxLinuxAdditions.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.1.4 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
Removing installed version 7.1.4 of VirtualBox Guest Additions...
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel 5.4.0-100-generic.

This system is currently not set up to build kernel modules.
Please install the gcc make perl packages from your distribution.
VirtualBox Guest Additions: Running kernel modules will not be replaced until
the system is restarted or 'rcvboxadd reload' triggered
VirtualBox Guest Additions: reloading kernel modules and services
VirtualBox Guest Additions: cannot reload kernel modules: one or more module(s)
is still in use
VirtualBox Guest Additions: kernel modules and services were not reloaded
VirtualBox Guest Additions: /log/vboxadd-setup.log may contain further information.
osboxes@osboxes:/mnt$ reboot
```

The terminal window also shows a sidebar with icons for a file manager, a help icon, and a terminal icon. The bottom of the window shows a taskbar with various system icons and the text 'Right Ctrl'.

done.